1 Theory shape relative to a reasoner

The axioms of a mechanics textbook do not yield the solutions of its exercises by a simple mechanical procedure: the reader brings methods, lemmas and understanding that are not written down.[1] The inducers of the previous sections cannot express this: they charge a theory by the length of a program for it, a short program can contain a generic solver, and by Craig's construction every consistent assigner of truth values is an axiom set at constant cost (**????**). A sense in which a written theory is well or badly shaped must be relative to a reasoner that uses it and to a cost that charges what is written. In the simplest model in which this can be stated, program length sees neither the reasoner nor the vocabulary; charged as written, the size of a usable theory depends exponentially on the pair (reasoner, vocabulary) (Table 1); and declarative against procedural form matters only by a constant factor once intermediate concepts are available and the reasoner can chain through them, and, without intermediate concepts or with a single round, by the number m of query directions of one parity law.

1.1 The model

Laws constrain the joint values of m observed quantities, and questions give some values and ask for another (the conditional questions of **??**, with literals as sentences).

Definition 1.1 (Theories, laws, reasoners, inducers)**.** Fix base variables $x_1, \dots, x_m$ and auxiliary variables $y_1, y_2, \dots$, the *intermediate concepts* (0-ary relation symbols of a first-order language $\mathcal{L}_m$ with equality). (a) An *explicit theory* Φ is a finite set of non-tautological clauses, a *linear theory* a finite set of equations over GF(2), in these variables, up to renaming of the auxiliary ones; $\operatorname{proj}\Phi := \{x \in \{0,1\}^m : \exists y\ (x,y) \models \Phi\}$. The hypothesis classes are $\mathcal{E}_m^{\text{base}}$ (no auxiliary variables) and $\mathcal{E}_m$ (any); likewise for linear theories. $|\Phi|$ counts clauses (equations), the *written length* $\|\Phi\|$ literal occurrences. (b) A *law* is a nonempty $\mathcal{S} \subseteq \{0,1\}^m$. $\mathcal{Q}(\mathcal{S})$ is the set of conditional questions (α, ℓ), α a partial assignment to the base variables consistent with $\mathcal{S}$ and ℓ a literal on a base variable outside $\operatorname{dom}\alpha$, with $\mathcal{S} \wedge \alpha \models \ell$; a *conditional data set* is a finite set of such pairs. (c) A *reasoner* $\mathcal{R}$ maps a theory and an antecedent α to a set of base literals or to a conflict $\perp$; it is *sound* if $\ell \in \mathcal{R}(\Phi, \alpha)$ implies $\Phi \cup \alpha \models \ell$, and $\perp$ occurs only if $\Phi \cup \alpha$ is unsatisfiable. The named reasoners are sound: UP_k, k *parallel* rounds of unit propagation (in each round every clause whose literals are all falsified but one, that one unassigned, assigns it; opposite derivations or a falsified clause give $\perp$); $\mathrm{UP} := \mathrm{UP}_\infty$; Gaussian elimination GE on linear theories and unbounded proof search PS_∞, which return exactly the entailed base literals. (d) Φ is $\mathcal{R}$-*compatible* with D if for every $(\alpha, \ell) \in D$, $\Phi \cup \alpha$ is satisfiable and $\ell \in \mathcal{R}(\Phi, \alpha)$; $\mathcal{R}$-*usable* for $\mathcal{S}$ if $\operatorname{proj}\Phi = \mathcal{S}$ and Φ is $\mathcal{R}$-compatible with $\mathcal{Q}(\mathcal{S})$. (e) The inducers range over the $\mathcal{R}$-compatible $\Phi \in \mathcal{E}_m$ (linear theories for GE); a superscript base restricts them to $\mathcal{E}_m^{\text{base}}$. *Program length*: $K_A^{\mathcal{R}}(D) := \min|p|$ over the p with $V(p) = \langle\Phi\rangle$, and $W_A^{\mathcal{R}}(D) := \sum 2^{-|p|}$ over the same p. *Listing*: $K_{\text{lst}}^{\mathcal{R}}(D) := \min \ell_{\text{lst}}(\Phi)$ and $W_{\text{lst}}^{\mathcal{R}}(D) := \sum 2^{-\ell_{\text{lst}}(\Phi)}$, for the prefix code $\ell_{\text{lst}} \geq \|\cdot\|$ of Equation (1), which lists the clauses literal by literal. *Time-charged*: $Kt_A^{\mathcal{R}}(D) := \min(|p| + \log t_p)$ over the p of $K_A^{\mathcal{R}}$, t_p the running time of V on p.

Under the two-valued encoding a single point of the law would fit all its question data (Proposition A.3). Under (E3), with unbounded proof search, the program-length inducers are K_A and W_A on question data, up to a constant for r.e. hypotheses and up to $m + O(\log m)$ bits for finite explicit theories (Proposition A.4): they extend SAI. Call $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$, with

[1] That written theories are far from sufficient for a simple reasoner, and are aids for a reasoner with much tacit understanding, is argued in Hänni's working notes (Hänni, 2026).

$A \in \mathrm{GF}(2)^{r\times m}$ of rank $r \geq 1$, an *affine law*. For affine laws and for the implication chain on $m \geq 3$ variables, compatibility with $\mathcal{Q}(\mathcal{S})$ already forces $\operatorname{proj}\Phi = \mathcal{S}$ (Lemma A.2).

1.2 Program length sees neither the reasoner nor the vocabulary

The prime implicates $\mathrm{PI}(\mathcal{S})$ of a law are its inclusion-minimal implicates among the clauses over the base variables; one round of unit propagation on them answers every question of $\mathcal{Q}(\mathcal{S})$ (Lemma A.5).

Theorem 1.2 (Reasoner invariance under program length)**.** *Let $\mathcal{R}'$ be a sound reasoner whose hypotheses are explicit or linear theories, of either class. For every conditional data set D, $K_A^{\mathrm{UP}_1,\mathrm{base}}(D) \leq K_A^{\mathcal{R}'}(D) + c$ and $W_A^{\mathrm{UP}_1,\mathrm{base}}(D) \geq 2^{-c} W_A^{\mathcal{R}'}(D)$; and $K_A^{\mathrm{UP}_1} = K_A^{\mathrm{UP}} = K_A^{\mathrm{PS}_\infty} \pm c$ for finite explicit theories of either class.*

Proof idea (Appendix A.2). If Φ' is $\mathcal{R}'$-compatible with D, soundness makes $\mathrm{PI}(\operatorname{proj}\Phi')$ UP_1-compatible with D, and a fixed prefix computes it from a program for Φ'.

This is Craig's collapse again, now for reasoners and vocabularies: the know-how a weak reasoner lacks can live in a generator of the written theory, at constant cost. It becomes visible when what is written, or the time to write it, is charged.

Theorem 1.3 (Where the reasoner and the vocabulary become visible)**.** *Let $\mathcal{S}$ be an affine law, $\mathcal{R}$ a sound reasoner, and take either class. (a) $K_{\mathrm{lst}}^{\mathcal{R}}(\mathcal{Q}(\mathcal{S}))$ is the least $\ell_{\mathrm{lst}}(\Phi)$ over the $\mathcal{R}$-usable Φ of the class. (b) $Kt_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S})) \geq \log\min\|\Phi\|$ over the same Φ; and $Kt_A^{\mathrm{UP}_1}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + \log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(r + \log m)$, $Kt_A^{\mathrm{GE}}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(\log m)$. (c) $K_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S})) = K(\mathcal{S}) \pm c$ for $\mathcal{R} \in \{\mathrm{UP}_k\ (k \geq 1), \mathrm{UP}, \mathrm{PS}_\infty, \mathrm{GE}\}$.*

Proof idea (Appendix A.2). (a) is Lemma A.2. (b) Printing Φ takes $\|\Phi\|$ steps; a program can compute (A,b) and print the prime implicates, a library with intermediate concepts, or the equations. (c) $\mathcal{S}$ gives $\mathrm{PI}(\mathcal{S})$, and a compatible Φ gives $\operatorname{proj}\Phi = \mathcal{S}$. So the separations below concern written size, which by (a) is K_{lst} up to the code length per literal.

1.3 Over the base variables: the exact minimum

How large must a theory be for unit propagation if it may use only the observed quantities? A *derived equation* of an affine law is $u = yA$, with $u \cdot x = b_u := y \cdot b$ on $\mathcal{S}$; d_* is the least weight of a nonzero one; a *circuit* is a nonzero derived equation whose support contains the support of no other. Let $C_{Z,\beta}$ be the clause over the variable set Z falsified exactly by the assignment β; the prime implicates are the $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ (Proposition A.7). A circuit u is *decomposable* if $u = u' + u''$ for nonzero derived equations whose supports meet in a single base variable outside $\mathrm{supp}(u)$: two derived equations chained through one intermediate base quantity. Let $\Phi_{\mathrm{ind}}(\mathcal{S}) := \{C_{\mathrm{supp}(u),\beta} : u \text{ an indecomposable circuit}, u \cdot \beta \neq b_u\}$.

Proposition 1.4 (What is forced)**.** *For every affine law, whatever d_*, the clauses contained in every* UP*-usable explicit theory over the base variables are exactly those of $\Phi_{\mathrm{ind}}(\mathcal{S})$.*

Theorem 1.5 (The unique minimum when $d_* \geq 3$)**.** *If $d_* \geq 3$, $\Phi_{\mathrm{ind}}(\mathcal{S})$ is* UP*-usable. Hence an explicit theory over the base variables equivalent to $\mathcal{S}$ is* UP*-usable if and only if it contains $\Phi_{\mathrm{ind}}(\mathcal{S})$, which is the unique minimum, with $\sum_{u\ \mathit{indecomposable}} 2^{|u|-1}$ clauses.*

Proof idea (Appendix A.3). From an antecedent that fixes all of an indecomposable circuit but one variable, the first round must fire a clause supported on that circuit, since a clause through another variable would decompose it; any other implicate can be deleted from the set of all implicates. For sufficiency, induct on the weight: a decomposable circuit is answered in two steps through its intermediate variable, and $d_* \geq 3$ makes both parts lighter.

So a circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one base quantity: base quantities already act as intermediate concepts (Corollary A.11 says when the given equations suffice; Remark A.12 treats $d_* \leq 2$).

Corollary 1.6 (The simplex family)**.** *Let $r \geq 3$, $m = 2^r - 1$, and let the columns of A be the nonzero vectors of* $\mathrm{GF}(2)^r$*. Every nonzero derived equation has weight $(m+1)/2$ and is an indecomposable circuit; the unique minimum* UP*-usable theory over the base variables is* $\mathrm{PI}(\mathcal{S})$*, with $m\,2^{(m-1)/2}$ clauses of width $(m+1)/2$, while* GE *uses the $r = \log(m+1)$ equations.*

The same holds for one round with any intermediate concepts (Proposition 1.7), but not for unbounded propagation with them: for $r \geq 4$, writing each circuit by its own chain of partial sums gives a UP-usable theory of $m(2m-4)$ clauses of width at most 3 (Remark A.15). So the simplex family separates the reasoners exponentially only over the base variables or for one round.

1.4 Intermediate concepts help only a reasoner that can chain through them

One parity law $\mathrm{Par}_m^b := \{x : x_1 \oplus \cdots \oplus x_m = b\}$, $m \geq 3$, needs all its 2^{m-1} clauses for unit propagation over the base variables (Corollary A.11). With the partial sums $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ as intermediate concepts, $4m-6$ clauses suffice for unit propagation when $m \geq 4$ (Proposition A.13), but only for a reasoner that can chain.

Proposition 1.7 (One round cannot use intermediate concepts)**.** *For any law $\mathcal{S}$, every* UP_1*-usable explicit theory, with any auxiliary variables, contains* $\mathrm{PI}(\mathcal{S})$*: 2^{m-1} clauses for one parity law, $m2^{(m-1)/2}$ for the simplex family.*

Proof idea (Appendix A.4). In one round from base literals, a clause with an auxiliary literal can derive only that literal, so the answers come from the clauses over the base variables, which are implicates of $\mathcal{S}$; a prime implicate can then be answered only by itself.

Proposition 1.8 (Rounds and size for one parity law)**.** *(a) For $m \geq 4$ the chain encoding is* UP_{m-1}*-usable and not* UP_{m-2}*-usable. (b) The* balanced-tree encoding*, along a rooted binary tree of height $\lceil \log m \rceil$ whose leaves are the base variables, with $y_v = y_{v'} \oplus y_{v''}$ (four clauses) at each internal node v other than the root and $y_{v'} \oplus y_{v''} = b$ (two clauses) at the root, has $4m-6$ clauses ($m \geq 2$) and is* $\mathrm{UP}_{2\lceil \log m \rceil - 1}$*-usable. (c) There is an absolute constant $c_{\mathrm{H}} > 0$ such that for every $k \geq 1$ and every m large enough given k, every* UP_k*-usable Φ for Par_m^b, with any auxiliary variables, has $3\|\Phi\| + 2m \geq 2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$; likewise, with $w-1$ for $m-1$, for every affine law with a circuit of weight w. (d) Hence* UP_k*-usable theories of size polynomial in m for one parity law need $k = \Omega(\log m / \log\log m)$; $2\lceil \log m \rceil - 1$ rounds suffice for $4m-6$ clauses.*

Proof idea (Appendix A.4). (a), (b): follow the round in which each auxiliary value becomes known. (c): k rounds of unit propagation are a monotone circuit of depth $2k$ with at most $k(3\|\Phi\| + 2m)$ gates (Lemma A.18) that computes the parity of $m-1$ bits; apply Håstad's lower bound for parity (Håstad, 1986).

reasoner $\mathcal{R}$	parity, base	parity, interm.	simplex, interm.	hidden state, interm.
UP_1	2^{m-1}	2^{m-1}	$m2^{(m-1)/2}$	$2^{\Omega(m^{\varepsilon/3})}$
UP_k, k fixed	2^{m-1}	$2^{\Omega(m^{1/(2k-1)})}$	$2^{\Omega(m^{1/(2k-1)})}$	$2^{\Omega(m^{\varepsilon/3})}$
UP_k, $k \geq 2\lceil \log m \rceil - 1$; UP	2^{m-1}	$\leq 4m-6$	$\leq m(2m-4)$	$2^{\Omega(m^{\varepsilon/3})}$
GE	1 equation	1 equation	$\log(m+1)$ equations	$\leq m$ equations

Table 1: The regimes (Cor. 1.10): the least size of a theory usable by the reasoner of the row, with the round budget the row states, in the class of the column (base: no auxiliary variables; interm.: any intermediate concepts). The inducer is $K^{\mathcal{R}}_{\mathrm{lst}}$ (superscript base in the first column), whose value on $\mathcal{Q}(\mathcal{S})$ is this size up to the code length per literal (Thm. 1.3(a)). Entries without a sign are exact and count clauses (equations for GE); $\leq$ marks the size of an explicit usable theory; $2^{\Omega(\cdot)}$ bounds the written length from below. Sources: first column Cor. A.11; UP_1 Prop. 1.7; fixed k and $4m-6$ Prop. 1.8; $m(2m-4)$ Rem. A.15; last column Thm. 1.9 (equations of weight ≤ 5). Under program length every entry is $K(\mathcal{S}) \pm c$ (Thm. 1.3(c)).

1.5 Systems of laws, and the regimes

For systems of laws, neither intermediate concepts nor depth need help. In the following law the observed quantities are fixed linear functionals of an unobserved state. Let Λ_q be the set of $v = (u, a) \in \mathrm{GF}(2)^{q+1}$ with $u \in \mathrm{GF}(2)^q$ of weight 1 or 3 and $a \in \{0, 1\}$, and $\mathbf{e}_\star := (0^q, 1)$. The *hidden-state law* on the $m = |\Lambda_q| + 1$ base variables $(x_v)_{v \in \Lambda_q}$, $x_\star$ is $\mathcal{H}_q := \{((z \cdot v)_{v \in \Lambda_q}, z \cdot \mathbf{e}_\star) : z \in \mathrm{GF}(2)^{q+1}\}$. With the state $z = (y, z_0)$ as intermediate concept it is written as the m equations $x_{(u,a)} = u \cdot y + a\, z_0$, $x_\star = z_0$, each of weight at most 5.

Theorem 1.9 (Systems of laws need exponential written length)**.** *The m equations above are* GE*-usable for $\mathcal{H}_q$. There are $\varepsilon > 0$ and q_0 such that for every $q \geq q_0$, every* UP*-usable explicit theory Φ for $\mathcal{H}_q$, with any auxiliary variables, clause widths and number of rounds, has $\|\Phi\| \geq 2^{q^\varepsilon/2 - O(1)} - m = 2^{\Omega(m^{\varepsilon/3})}$.*

Proof idea (Appendix A.4). Setting a set S of the x_v to 0 determines $x_\star$ exactly when the XOR system $\{u \cdot y = a : (u, a) \in S\}$ is unsatisfiable. So unit propagation on a usable Φ, read as a monotone circuit (Lemma A.18), decides this with $O((\|\Phi\| + m)^2)$ gates; the monotone 3-XOR-SAT function of Göös et al. (2019) is a projection of it and needs monotone circuits of size 2^{q^ε}.

The proof is a short combination of the monotone-circuit view of unit propagation (Bessiere et al., 2009) with the bound of Göös et al. (2019); it improves the superpolynomial bound of Gwynne and Kullmann (2014) for arc-consistent representations of XOR systems, which are our usable theories (Lemma A.17 and Remark A.19). We did not find it stated; it is probably known to specialists. Intermediate concepts private to one law add nothing to a system (Proposition A.16). A solver whose unit propagation does linear algebra over parity constraints (Beame and Sun, 2026) is a reasoner for which such laws are cheap.

Corollary 1.10 (The regimes)**.** *For one parity law ($m \geq 3$), the simplex family and the hidden-state law, the least sizes of theories usable by* UP_1*, by* UP_k *(k fixed, or $k \geq 2\lceil \log m \rceil - 1$), by* UP *and by* GE*, over the base variables or with any intermediate concepts, are as stated in Table 1.*

So the exponential separations are between pairs (reasoner, vocabulary). Intermediate quantities help a reasoner only if it can chain through them: not at all with one round, not enough

with any fixed number of rounds, and fully, for one law, from about $2 \log m$ rounds. For the hidden-state law no vocabulary helps unit propagation at any depth, while elimination needs only the natural one.

1.6 Directionality

A natural guess is that logical form matters because a law can be used in every direction (Kepler's third law gives the period from the semi-major axis and conversely), while a program computes one. Take as the procedural counterpart a *rule base*: a set of sound directed rules $P \Rightarrow \ell$ over the base variables (a set P of literals consistent with $\mathcal{S}$ and a literal ℓ on another variable, with $\mathcal{S} \wedge P \models \ell$), used by forward chaining. For one parity law the only usable rule base is the set of all $m2^{m-1}$ sound rules, m times the 2^{m-1} clauses of the least UP-usable theory over the base variables: a factor m, the number of query directions, and the same factor holds for one round of reasoning with any vocabulary. But for every law and round budget the factor is at most the largest clause width of a smallest usable theory, with intermediate concepts and unbounded rounds it is at most 9 in written length, a program with internal variables answers all m directions of parity with $O(m)$ operations, and under program length the factor vanishes (Theorem A.21). So the dependence on directions is real but cheap: logical form does not get its support from directionality.

1.7 Where the know-how lives

The know-how to answer every question can be paid for in the reasoner (the r equations with GE), in written lemmas, or in a computation that the cost model does not charge: a generator under program length (Theorem 1.2), or a membership test with free search for derivations. For the latter, $\mathrm{PI}(\mathcal{S})$ has a decider of length $K^q(A, b) + c$ (q a suitable polynomial) running in time polynomial in m, and each $\bigwedge \alpha \to \ell$ with $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$ follows from one of its members by propositional reasoning (Proposition A.24). Under the readings of the charged deduction of **??**, extended to conditional data (Definition A.25), this decider is a compatible hypothesis of weight at least $2^{-K^q(A,b)-O(\log m)}$, given a polynomial bound on the derivations that we sketch, while every compatible hypothesis has weight at most $2^{-K(\mathcal{S})+c}$ (Theorem A.26). So these readings give the full lemma library ($m2^{(m-1)/2}$ clauses for the simplex family) a cost within $K^q(A, b) - K(\mathcal{S}) + O(\log m)$ bits of the least possible: the know-how lives in the decider, which runs elimination. This locates the fit classes of **??**. The readings bound the size of a derivation that exists and do not charge finding it (**??**); they model a reasoner that is handed the derivation, and cannot express the textbook point, which concerns a reasoner that must find the answer. A deterministic bounded reasoner with a cost on the written theory can.

1.8 Learning, and depth against size

A weak reasoner must also acquire its know-how (Theorem A.29). For one parity law ($m \geq 3$), call the non-solution obtained by flipping the asked variable in the completion the *pattern* of a question; there are 2^{m-1} patterns. Over the base variables, every inclusion-minimal UP-compatible theory fails on some question of every unseen pattern, so it is fully correct only after all have been seen, and on data made of whole patterns the listing inducer pays at least $\mathrm{len}_m \approx m \log m$ bits per pattern (semimeasure convention); with GE, or with intermediate concepts and unbounded propagation, it pays $O(m \log m)$ bits in all, and under program length there is no gap.

Depth trades against size even without new vocabulary (Theorem A.31). For the implication chain $x_1 \to \cdots \to x_m$, the least UP_k-usable theory over the base variables has as many clauses as a minimum k-hop transitive-closure spanner of the directed path, $\Theta(m\,\mathrm{ia}_k(m))$ (Alon and Schieber, 1987; Bhattacharyya et al., 2012), ia_k the k-th inverse-Ackermann row: $\binom{m}{2}$ at $k = 1$, $\Theta(m \log m)$ at $k = 2$, down to the $m - 1$ axioms of the law at $k \geq m - 1$. At depth 2, intermediate concepts and wide clauses save at most a factor of about 2 (Theorem A.34). In first-order logic the cut-free proofs from a lemma-free theory can be non-elementarily longer (Statman, 1979; Orevkov, 1982), and we sketch how valid lemma sentences restore short ones (Remark A.37).

1.9 What this shows and does not show

Question 1.11. Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws: to real-valued laws with numerical solvers, to probabilistic programs with conditioning, to simulation codes?

This is the precise form of the open question why theories are shaped like logical theories (**??**).

Reading. Whether a written theory suffices depends on the reader. Paired with a reasoner that can do algebra, a linear law's few equations suffice. Paired with one that only applies rules, the theory must spell out exponentially many derived laws, unless it is written with new intermediate quantities and the reader can chain many steps through them. Even then, some systems of laws need exponentially long written theories for such a reader, whatever quantities are introduced. The idealised inducer cannot see any of this, because it charges a theory by the length of a program that may compute the missing lemmas. The model does not show that logic, rather than probabilistic programs with conditioning or classes of models, is the right formal home for theories: the large separations are between pairs (reasoner, vocabulary), and a rule base with intermediate concepts is within a constant factor of a clausal theory. It does not touch the gap between a problem as posed and its formalisation, nor tacit skill (**??**). Its laws are linear or implicational, its reasoners unit propagation, forward chaining and elimination, and its cost the written length. It is not a theorem about textbooks.

A Proofs for theory shape

Throughout, $\mathcal{S}$ is a law on the base variables $x_1, \ldots, x_m$, Φ an explicit or linear theory with m_{aux} auxiliary variables and $\bar{m} := m + m_{\mathrm{aux}}$ variables in all, and $\mathbf{e}_j$ the j-th unit vector. A partial assignment is identified with the set of literals it makes true, and $\hat{x}$ is the conjunction of the base literals true at $x \in \{0,1\}^m$. For a sentence or literal φ, $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

A.1 The model: the listing code, the two encodings, the bridge

Theories that differ only by a renaming of their auxiliary variables are identified, and each class has a fixed canonical representative whose auxiliary variables are $y_1, \ldots, y_{m_{\mathrm{aux}}}$ and whose clauses are listed in a fixed canonical order (for instance the lexicographically least listing). Let $\ell_{\mathbb{N}}(n) := |\mathrm{El}(n+1)|$ for $n \geq 0$, with El the Elias delta code of the positive integers; $\ell_{\mathbb{N}}$ is nondecreasing and the lengths of a prefix code of $\{0, 1, 2, \ldots\}$. The *listing length* of a canonical explicit theory is

$$\ell_{\mathrm{lst}}(\Phi) := \ell_{\mathbb{N}}(m_{\mathrm{aux}}) + \ell_{\mathbb{N}}(|\Phi|) + \sum_{C \in \Phi} \big(\ell_{\mathbb{N}}(|C|) + |C|(\lceil \log \bar{m} \rceil + 1)\big), \tag{1}$$

each literal being written as a variable index and a sign; a linear theory is listed likewise, each equation by its support and its constant, with $\ell_{\mathbb{N}}(|u|) + |u|\lceil \log \bar{m} \rceil + 1$ bits for an equation of support size $|u|$. Clearly $\ell_{\mathrm{lst}}(\Phi) \geq \|\Phi\|$.

Lemma A.1 (The listing code)**.** $\sum 2^{-\ell_{\mathrm{lst}}(\Phi)} \leq 1$ *over the canonical explicit theories, and* $\ell_{\mathrm{lst}}(\Phi \setminus \{C\}) < \ell_{\mathrm{lst}}(\Phi)$ *for every* $C \in \Phi$*. In particular an* ℓ_{lst}*-shortest* $\mathcal{R}$*-compatible theory is inclusion-minimal among the* $\mathcal{R}$*-compatible theories.*

Proof. Fix m_{aux}. The code of a single clause (its length by $\ell_{\mathbb{N}}$, then that many literals of $\lceil \log \bar{m} \rceil + 1$ bits each) is a prefix code on finite sequences of literals, so $\sum_C 2^{-\ell(C)} \leq 1$ over clauses written in canonical order, where $\ell(C)$ is the summand of (1). The sets of s distinct clauses have total weight at most $(\sum_C 2^{-\ell(C)})^s / s! \leq 1$. The codes $\ell_{\mathbb{N}}(m_{\mathrm{aux}})$ and $\ell_{\mathbb{N}}(s)$ contribute factors whose sums over m_{aux} and over s are at most 1, and canonical theories are a subset of all clause sets; this gives the Kraft inequality. Removing a clause C removes $\ell(C) > 0$ bits and replaces $\ell_{\mathbb{N}}(s)$ by $\ell_{\mathbb{N}}(s-1) \leq \ell_{\mathbb{N}}(s)$; if it removes the last occurrence of an auxiliary variable, then m_{aux} and $\lceil \log \bar{m} \rceil$ do not increase either (after renaming to canonical form, which does not change ℓ_{lst}). □

For two base variables the Kraft sum is 0.2525 (a direct computation over all 256 theories; code in the accompanying repository).

Lemma A.2 (Compatibility with all questions forces the projection)**.** *Let* $\mathcal{R}$ *be sound and* Φ $\mathcal{R}$*-compatible with* $\mathcal{Q}(\mathcal{S})$*. Then* $\operatorname{proj} \Phi \subseteq \mathcal{S}$*; and* $\operatorname{proj} \Phi = \mathcal{S}$ *if every* $x \in \mathcal{S}$ *has a coordinate that is determined on* $\mathcal{S}$ *by the others. This holds for every affine law and for the chain law* $\mathrm{Ch}_m := \{x : x_1 \leq \cdots \leq x_m\}$ *with* $m \geq 3$*; so for these laws,* $\mathcal{R}$*-compatibility with* $\mathcal{Q}(\mathcal{S})$ *and* $\mathcal{R}$*-usability coincide.*

Proof. Let $x \notin \mathcal{S}$. Choose $I \subsetneq [m]$ maximal such that $\alpha := x|_I$ is consistent with $\mathcal{S}$ ($I = \emptyset$ qualifies since $\mathcal{S} \neq \emptyset$, and $I \neq [m]$ since $x \notin \mathcal{S}$). For $j \notin I$, the extension of α by the value of x_j at x is inconsistent with $\mathcal{S}$, so $\mathcal{S} \wedge \alpha$ entails the literal ℓ on x_j that is false at x, and $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$. Compatibility and soundness give $\Phi \cup \alpha \models \ell$; x extends α and falsifies ℓ, so $x \notin \operatorname{proj} \Phi$. Conversely, let $x \in \mathcal{S}$ have its coordinate j determined by $\alpha := x|_{[m]\setminus\{j\}}$. Then $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$ for the literal ℓ on x_j true at x, so $\Phi \cup \alpha$ is satisfiable, by a point of $\operatorname{proj} \Phi \subseteq \mathcal{S}$ extending α, and the only such point of $\mathcal{S}$ is x. For an affine law, a coordinate j is determined by the others if and only if j lies in the support of a nonzero derived equation (Lemma A.6 with $I = [m] \setminus \{j\}$); such a j exists since $r \geq 1$, and the property does not depend on x. For Ch_m, the points are $0^a 1^{m-a}$; if $a \geq 2$, then $x_2 = 0$ forces $x_1 = 0$, and if $m - a \geq 2$, then $x_{m-1} = 1$ forces $x_m = 1$; one of the two holds as $m \geq 3$. □

The hypothesis $m \geq 3$ is needed: $\Phi = \{\neg x_1 \vee x_2,\ x_1 \vee \neg x_2\}$ is UP- and PS_∞-compatible with $\mathcal{Q}(\mathrm{Ch}_2)$, whose only questions are "$x_1 = 1$, so $x_2 = 1$" and "$x_2 = 0$, so $x_1 = 0$", but $\operatorname{proj} \Phi = \{00, 11\} \neq \mathrm{Ch}_2$; the point 01 has no determined coordinate.

Proposition A.3 (The two-valued encoding is degenerate)**.** *Encode a conditional datum* (α, ℓ) *by the sentence* $\bigwedge \alpha \to \ell$ *labelled true, with* $B = \emptyset$*. Then every consistent theory that proves all sentences proved by a compatible theory is compatible. In particular, for every law* $\mathcal{S}$ *and every* $x^0 \in \mathcal{S}$*, the theory* $T_{x^0} := \{x_i^{x^0_i} : i \leq m\}$ *proves* $\bigwedge \alpha \to \ell$ *for every* $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$*, so* $K_A(\{(\bigwedge \alpha \to \ell, 1) : (\alpha, \ell) \in \mathcal{Q}(\mathcal{S})\}) \leq K(x^0) + c$*; for a homogeneous affine law (*$b = 0$*,* $x^0 = 0$*) this is at most* $K(m) + c$*.*

Proof. Compatibility with a datum labelled true asks only that the theory be consistent and prove the sentence, which passes to consistent theories proving more. If α agrees with x^0, then x^0 is a point of $\mathcal{S}$ extending α, so x^0 satisfies ℓ and $T_{x^0} \vdash \ell$; otherwise $T_{x^0} \vdash \neg \bigwedge \alpha$. Either way $T_{x^0} \vdash \bigwedge \alpha \to \ell$, and T_{x^0} is consistent and computable from x^0. □

Under this encoding the data cannot distinguish a law from a single world that satisfies it. Encoding (E3) of **??** makes "given α" hypothetical: a compatible theory may not refute an antecedent, so no single world fits question data with two incompatible antecedents. (E3) is not upward closed, so it cannot be expressed by two-valued proof-requiring labels. For a conditional data set D we write $K_A(D)$ and $W_A(D)$ for the inducers of **??** over the language $\mathcal{L}_m$ with $B = \emptyset$, in which an r.e. theory T is compatible with D if it is compatible with every datum of D under (E3).

Proposition A.4 (The bridge to SAI)**.** *(a) With $\mathcal{R} = \mathrm{PS}_\infty$ and as hypotheses the r.e. sets of $\mathcal{L}_m$-sentences, or the r.e. sets of clauses over the base variables, $K_A^{\mathrm{PS}_\infty}(D) = K_A(D) \pm c$ and $W_A^{\mathrm{PS}_\infty}(D) \asymp W_A(D)$ on every conditional data set D. (b) With finite explicit theories, of either class, $K_A(D) - c \le K_A^{\mathrm{PS}_\infty}(D) \le K_A(D) + \ell_{\mathbb{N}}(m) + \ell_{\mathbb{N}}(2^m) + c$, that is, up to $m + O(\log m)$ bits.*

Proof. (a) For sets of sentences, PS_∞-compatibility of an r.e. set T is (E3), by the completeness theorem: $T \cup \alpha$ is satisfiable if and only if it is consistent, and PS_∞ returns ℓ if and only if $T \cup \alpha \vdash \ell$. For sets of clauses, a fixed program turns an enumeration of T into an enumeration of $\mathrm{Cl}(T)$, the set of clauses over the base variables derivable from T. For every question, $T \cup \alpha$ is inconsistent iff $T \vdash \neg \bigwedge \alpha$ iff the clause $\neg\alpha$ is in $\mathrm{Cl}(T)$ iff $\mathrm{Cl}(T) \cup \alpha$ is inconsistent (as $\mathrm{Cl}(T) \subseteq \mathrm{Cn}(T)$); and $T \cup \alpha \vdash \ell$ iff the clause $\neg\alpha \vee \ell$ (non-tautological, as ℓ is on a variable outside $\operatorname{dom}\alpha$) is in $\mathrm{Cl}(T)$ iff $\mathrm{Cl}(T) \cup \alpha \vdash \ell$. So T and $\mathrm{Cl}(T)$ are compatible with the same conditional data. (A theory of $\mathcal{L}_m$ need not consist of Boolean combinations of the atoms, since $\mathcal{L}_m$ has equality and quantifiers; $\mathrm{Cl}(T)$ replaces a normal form.) Conversely, clauses are sentences. The weights compare in the same way, through the fixed maps on programs. (b) *Left.* A fixed prefix replaces a finite explicit theory Φ, whose auxiliary variables are not symbols of $\mathcal{L}_m$, by $\mathrm{PI}(\operatorname{proj}\Phi)$, a finite set of $\mathcal{L}_m$-sentences. It is compatible with the same data: $\Phi \cup \alpha$ is satisfiable iff $\operatorname{proj}\Phi$ meets α, and for a base literal ℓ, $\Phi \cup \alpha \models \ell$ iff every point of $\operatorname{proj}\Phi$ extending α satisfies ℓ. *Right.* Let p enumerate a consistent T that is compatible with D under (E3), and let $\mathcal{S}_T := \{x : T \cup \{\hat{x}\} \text{ consistent}\}$. Then $\mathcal{S}_T \neq \emptyset$, since otherwise T proves $\bigwedge_x \neg\hat{x}$, which is refutable. Since $x \notin \mathcal{S}_T$ iff $T \vdash \neg\hat{x}$, a program given p, m and $|\mathcal{S}_T|$ (self-delimiting, with $\ell_{\mathbb{N}}(m) + \ell_{\mathbb{N}}(|\mathcal{S}_T|) + c$ extra bits) enumerates refutations until $2^m - |\mathcal{S}_T|$ points are refuted, and outputs $\mathrm{PI}(\mathcal{S}_T)$. By completeness, $T \cup \alpha$ is consistent iff $\mathcal{S}_T$ meets α, and $T \cup \alpha \vdash \ell$ iff every point of $\mathcal{S}_T$ extending α satisfies ℓ, as ℓ is a base literal. So $\mathrm{PI}(\mathcal{S}_T)$ is PS_∞-compatible with D exactly when T is, and $\ell_{\mathbb{N}}(|\mathcal{S}_T|) \le \ell_{\mathbb{N}}(2^m)$. □

So the reasoner-relative inducers extend SAI: with an ideal deducer and (E3) they are SAI on question data, up to the gap between finite and r.e. hypotheses in (b). We do not know whether that gap is attained.

A.2 Program length, listing and time

Lemma A.5 (Prime implicates and one round of propagation)**.** *Let $\mathcal{S}$ be any law. (a) If α is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge \alpha \models \ell$, one round of unit propagation on $\mathrm{PI}(\mathcal{S}) \cup \alpha$ derives ℓ. (b) Every UP_1-usable $\Phi \in \mathcal{E}_m^{\mathrm{base}}$ contains $\mathrm{PI}(\mathcal{S})$.*

Part (a) is folklore; compiling a theory so that unit resolution becomes complete is studied by del Val (1994).

Proof. (a) The clause $\neg\alpha \vee \ell$ is an implicate of $\mathcal{S}$, so it contains a prime implicate C. If $\ell \notin C$, then $C \subseteq \neg\alpha$ is falsified by α, contradicting the consistency of α with $\mathcal{S}$. So $\ell \in C$ and all other literals of C are falsified by α: C fires in round 1. No conflict arises, since $\mathrm{PI}(\mathcal{S}) \cup \alpha$ is satisfiable. (b) Let $C = \neg\alpha' \vee \ell \in \mathrm{PI}(\mathcal{S})$. Since C is prime, $\neg\alpha'$ is not an implicate, so α' is consistent with $\mathcal{S}$ and $(\alpha', \ell) \in \mathcal{Q}(\mathcal{S})$. In one round from α', the clause of Φ that derives ℓ has all its other literals falsified by α', so it is some $C' \subseteq C$ with $\ell \in C'$; C' is an implicate of $\mathcal{S}$ (as Φ is a theory over the base variables with $\operatorname{proj}\Phi = \mathcal{S}$), so $C' = C$ by primality. □

Proof of Theorem 1.2. Let $\mathrm{comp}(\Phi') := \mathrm{PI}(\operatorname{proj}\Phi')$, computable from Φ' (a linear theory is first written as clauses). If Φ' is $\mathcal{R}'$-compatible with D then, by soundness, for each $(\alpha, \ell) \in D$ the set $\Phi' \cup \alpha$ is satisfiable and entails ℓ; so $\operatorname{proj}\Phi'$ meets α, and every point of it extending α satisfies ℓ, as ℓ is a base literal. By Lemma A.5(a) applied to the law $\operatorname{proj}\Phi'$, $\mathrm{comp}(\Phi') \in \mathcal{E}_m^{\mathrm{base}}$ is UP_1-compatible with D. A program for $\mathrm{comp}(\Phi')$ is a fixed prefix followed by a program for Φ', which gives both inequalities. For the equalities: UP_1, UP and PS_∞ are sound, so each is covered by the inequality, within either class; conversely a UP_1-compatible theory is UP- and PS_∞-compatible, since on a satisfiable $\Phi \cup \alpha$ these reasoners derive at least what UP_1 derives and never reach a conflict. □

On the question data of any law, a one-round unit-propagation reasoner over the base variables is therefore as good, under program length, as Gaussian elimination or unbounded proof search with any intermediate concepts, up to a machine constant.

Proof of Theorem 1.3. (a) By Lemma A.2, for an affine law and a sound reasoner, compatibility with $\mathcal{Q}(\mathcal{S})$ is usability. (b) The lower bound is the printing time: V needs at least $\|\Phi\|$ steps to print $\langle\Phi\rangle$. For the upper bounds, a program runs a program p for (A, b) (time t_p), enumerates the 2^r derived equations, keeps the circuits (time $\mathrm{poly}(m)4^r$), and prints either their clauses, which form $\mathrm{PI}(\mathcal{S})$ (Proposition A.7; time $\mathrm{poly}(m)|\mathrm{PI}(\mathcal{S})|$), or the library of Proposition A.14 with intermediate concepts (at most 2^r circuits, each with at most $4m$ clauses), or, for GE, just the r equations, without enumerating the derived equations. Then log of the total time is at most $\log t_p$ plus $\log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, resp. $O(r + \log m)$, resp. $O(\log m)$, and $|p| + \log t_p = Kt(A, b)$ for an optimal p. (A program of length $K(A, b)$ for (A, b) need not run within any computable time bound, which is why Kt appears.) (c) Upper bounds: from $\mathcal{S}$ a fixed program computes $\mathrm{PI}(\mathcal{S}) \in \mathcal{E}_m^{\mathrm{base}}$, which is usable for every UP_k, UP and PS_∞ (Lemma A.5(a)), and a canonical system of equations for $\mathcal{S}$, which is usable for GE. Lower bound: from a program for any compatible Φ a fixed program computes $\operatorname{proj}\Phi = \mathcal{S}$ (Lemma A.2). □

A.3 Linear laws over the base variables

Throughout this subsection $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$ with $A \in \mathrm{GF}(2)^{r\times m}$ of rank $r \geq 1$, $\mathrm{Row}(A)$ is the set of derived equations, $b_u := y \cdot b$ for $u = yA$ (well defined as A has full row rank, and linear in u), and a_* is the number of derived equations of weight d_*. Over GF(2) a vector is determined by its support, so a circuit is the unique nonzero derived equation supported in its support.

Lemma A.6 (Determination)**.** *For $I \subseteq [m]$ and $j \notin I$ the following are equivalent: (i) x_j is a function of x_I on $\mathcal{S}$; (ii) some $u \in \mathrm{Row}(A)$ has $j \in \mathrm{supp}(u) \subseteq I \cup \{j\}$; (iii) some circuit has this property.*

Proof. (ii)⇒(i): $x_j = b_u + \sum_{i\in\mathrm{supp}(u)\setminus\{j\}} x_i$ on $\mathcal{S}$. (i)⇒(ii): if no such u exists, then $\mathbf{e}_j \notin \mathrm{Row}(A) + \mathrm{span}\{\mathbf{e}_i : i \in I\}$, so some z is orthogonal to that space and has $z_j = 1$; then $z \in \ker A$ and $z_i = 0$

on I, and $x, x+z \in \mathcal{S}$ agree on I and differ at j. (ii)⇒(iii): take u of least support among the witnesses of (ii). If a nonzero derived equation u' had $\mathrm{supp}(u') \subsetneq \mathrm{supp}(u)$, then either $j \in \mathrm{supp}(u')$, contradicting minimality, or $u+u'$, with support $\mathrm{supp}(u) \setminus \mathrm{supp}(u') \ni j$, is a smaller witness. So u is a circuit. □

Proposition A.7 (Prime implicates of an affine law)**.** *For $Z \subseteq [m]$, the projection of $\mathcal{S}$ to Z is the affine subspace cut out by the derived equations supported in Z; so $C_{Z,\beta}$ is an implicate of $\mathcal{S}$ if and only if β violates some derived equation supported in Z. The prime implicates of $\mathcal{S}$ are exactly the clauses $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$; there are $\sum_{\text{circuits } u} 2^{|u|-1}$ of them, and $\mathrm{PI}(\mathcal{S})$ is UP_1-usable.*

Proof. An affine subspace is the common solution set of the affine functionals constant on it. The functionals on $\{0,1\}^Z$ that are constant on the projection of $\mathcal{S}$ are the restrictions of the functionals constant on $\mathcal{S}$, that is of $\mathrm{Row}(A)$ (as $\mathcal{S} = x^0 + \ker A$ and $(\ker A)^\perp = \mathrm{Row}(A)$), that vanish off Z. So $C_{Z,\beta}$ is an implicate iff β is outside the projection iff β violates some derived equation u with $\mathrm{supp}(u) \subseteq Z$. Every such u is a sum of circuits supported in $\mathrm{supp}(u)$: if u is not a circuit, some circuit u' has $\mathrm{supp}(u') \subsetneq \mathrm{supp}(u)$, and $u+u'$ has the smaller support $\mathrm{supp}(u) \setminus \mathrm{supp}(u')$; induct. Since $u \mapsto b_u$ is linear, β violates one of these circuits, u_0, and $C_{\mathrm{supp}(u_0),\beta|_{\mathrm{supp}(u_0)}} \subseteq C_{Z,\beta}$ is an implicate. A clause $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ is prime, since a strictly smaller implicate would be violated on a smaller support, by a derived equation supported there, against the minimality of circuits. For each circuit there are $2^{|u|-1}$ such β, and distinct circuits have distinct supports. $\mathrm{PI}(\mathcal{S})$ defines $\mathcal{S}$, and is UP_1-compatible with $\mathcal{Q}(\mathcal{S})$ by Lemma A.5(a). □

Proposition A.8 (Any CNF)**.** *Every implicate of $\mathcal{S}$ has width at least d_*, so every CNF over the base variables equivalent to $\mathcal{S}$ has at least $(1-2^{-r})2^{d_*}$ clauses.*

Proof. For $|Z| < d_*$ no nonzero derived equation is supported in Z, so the projection of $\mathcal{S}$ to Z is $\{0,1\}^Z$ (Proposition A.7) and no clause over Z is an implicate. A clause of width $w \geq d_*$ excludes $2^{m-w} \leq 2^{m-d_*}$ points, and the $2^m - 2^{m-r}$ non-solutions must all be excluded. □

Lemma A.9 (Subsumption)**.** *Let $\mathcal{S}$ be any law, Φ a set of implicates of $\mathcal{S}$ over the base variables, and Φ' obtained by replacing each clause C by any prime implicate $C' \subseteq C$. For every α consistent with $\mathcal{S}$ and every k, every literal derived by UP_k on $\Phi \cup \alpha$ is derived by UP_k on $\Phi' \cup \alpha$, and neither run reaches a conflict. So minimum usable theories over the base variables may be sought among subsets of $\mathrm{PI}(\mathcal{S})$, round for round.*

Proof. Both runs derive only literals entailed by $\mathcal{S} \wedge \alpha \neq \emptyset$, so neither conflicts. Induct on rounds. If C fires in round t of the Φ-run, deriving L, all other literals of C are falsified after round $t-1$, hence, by induction, in the Φ'-run. If $L \notin C'$, every literal of $C' \subseteq C$ would be falsified by the state of the Φ-run after round $t-1$; that state holds at every point of $\mathcal{S} \wedge \alpha$, and such a point would falsify the implicate C', which is impossible. So $L \in C'$, and C' fires by round t. □

Only one direction holds: Φ' derives at least what Φ derives, and the converse fails (see the counterexamples after Theorem A.31).

Decomposable circuits. Recall that a circuit u is decomposable if $u = u' + u''$ with $u', u'' \in \mathrm{Row}(A)$ nonzero and $\mathrm{supp}(u') \cap \mathrm{supp}(u'') = \{i\}$ for a single $i \notin \mathrm{supp}(u)$. Equivalently:

$$\begin{gathered}\text{for some, equivalently every, } j \in \mathrm{supp}(u) \text{ there are } u' \in \mathrm{Row}(A) \text{ and } i \notin \mathrm{supp}(u) \\ \text{with } i \in \mathrm{supp}(u') \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}.\end{gathered} \tag{2}$$

Indeed, a decomposition through i gives such a u' for every j, taking u' or u'', whichever does not contain j (they partition $\mathrm{supp}(u)$); conversely such a u' gives the decomposition $u = u' + (u + u')$, whose supports meet in $\{i\}$.

Proposition A.10 (Indecomposable circuits must be written)**.** *For every d_*, every* UP*-usable explicit theory over the base variables contains all $2^{|u|-1}$ clauses $C_{\mathrm{supp}(u),\beta}$, $u \cdot \beta \neq b_u$, of every indecomposable circuit u.*

Proof. Let Φ be usable (so its clauses are implicates), u indecomposable, $j \in \mathrm{supp}(u)$, $I := \mathrm{supp}(u) \setminus \{j\}$, and α any assignment to I. No nonzero derived equation is supported in I, as u is a circuit, so α is consistent and determines x_j (Proposition A.7 and Lemma A.6). Unit propagation never conflicts on the satisfiable $\Phi \cup \alpha$, and by usability it derives a literal outside I; so round 1 derives one. A literal derived in round 1 on x_i, $i \notin I$, comes from a clause $C \in \Phi$ whose other literals are on I and falsified by α, so $\mathrm{supp}(C) \subseteq I \cup \{i\}$, and C is an implicate. By Proposition A.7 some derived equation u' with $\mathrm{supp}(u') \subseteq \mathrm{supp}(C)$ is violated by the assignment that falsifies C, and $i \in \mathrm{supp}(u')$, since otherwise u' would be supported in I. If $i \neq j$, then $i \notin \mathrm{supp}(u)$ and u' contradicts indecomposability by (2). So $i = j$, $\mathrm{supp}(u') \subseteq \mathrm{supp}(u)$, hence $u' = u$ by the minimality of circuits, and $C = C_{\mathrm{supp}(u),\beta}$ for the assignment β that extends α by the value of x_j opposite to the derived one. The $2^{|u|-1}$ choices of α give all the clauses of u. □

Proof of Proposition 1.4. By Proposition A.10 it remains to show that for every clause $C \notin \Phi_{\mathrm{ind}}(\mathcal{S})$ some UP-usable theory over the base variables omits C. Let $\mathrm{Imp}(\mathcal{S})$ be the set of all non-tautological implicates over the base variables; we show that $\mathrm{Imp}(\mathcal{S}) \setminus \{C\}$ is usable. It consists of implicates, so unit propagation on it from a consistent antecedent never conflicts, and once it is compatible with $\mathcal{Q}(\mathcal{S})$ its projection is $\mathcal{S}$ (Lemma A.2). If C is not a prime implicate, the set contains $\mathrm{PI}(\mathcal{S})$ and is usable by Lemma A.5(a) and the monotonicity of unit propagation in the clause set. Otherwise $C = C_{\mathrm{supp}(u),\beta}$ for a decomposable circuit u. Let $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$. In the one-round run on $\mathrm{PI}(\mathcal{S}) \cup \alpha$ some prime implicate $C' \subseteq \neg\alpha \vee \ell$ fires. If $C' \neq C$, it is in our set. If $C' = C$, then ℓ is the literal of C on some $j \in \mathrm{supp}(u)$, and α contains $\beta|_{\mathrm{supp}(u)\setminus\{j\}}$. By (2) there are u' and $i \notin \mathrm{supp}(u)$ with $i \in \mathrm{supp}(u') \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$; put $u'' := u + u'$, so that $j \in \mathrm{supp}(u'') \subseteq (\mathrm{supp}(u) \setminus \mathrm{supp}(u')) \cup \{i\}$. Let v_i be the value of x_i that u' forces given α. The clause on $\mathrm{supp}(u')$ falsified by $\alpha|_{\mathrm{supp}(u')\setminus\{i\}} \cup \{x_i = 1 - v_i\}$ is an implicate (that assignment violates u'), differs from C (its support contains $i \notin \mathrm{supp}(u)$) and fires in round 1, deriving $x_i = v_i$. The clause on $\mathrm{supp}(u'')$ falsified by the restriction of $\alpha \cup \{x_i = v_i\}$ to $\mathrm{supp}(u'') \setminus \{j\}$, extended by the value of x_j opposite to ℓ, is likewise an implicate different from C, and fires in round 2, deriving ℓ ($\alpha \cup \{x_i = v_i\}$ is consistent and entails ℓ). So every question is answered. □

Proof of Theorem 1.5. We show by induction on $|w|$: for every nonzero $w \in \mathrm{Row}(A)$, every $j \in \mathrm{supp}(w)$ and every assignment α to $\mathrm{supp}(w) \setminus \{j\}$ consistent with $\mathcal{S}$, unit propagation on $\Phi_{\mathrm{ind}} \cup \alpha$ derives the value of x_j, and it never conflicts, as $\Phi_{\mathrm{ind}} \cup \alpha$ is satisfiable. By Lemma A.6 and the monotonicity of unit propagation in the antecedent (a larger consistent antecedent derives at least as much), this is usability; the projection is $\mathcal{S}$ by Lemma A.2.

- *w is not a circuit.* By Lemma A.6(iii) with $I = \mathrm{supp}(w) \setminus \{j\}$ there is a circuit $c' \neq w$ with $j \in \mathrm{supp}(c') \subseteq \mathrm{supp}(w)$, hence $|c'| < |w|$; apply the induction hypothesis to c' and the restriction of α.
- *$w = u$ is an indecomposable circuit.* Its clauses are in Φ_{ind}, and one of them fires.
- *$w = u$ is a decomposable circuit.* By (2) write $u = u' + u''$ through $i \notin \mathrm{supp}(u)$ with $j \in \mathrm{supp}(u'')$. If $\mathrm{supp}(u') = (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$, then $u'' = \mathbf{e}_i + \mathbf{e}_j$ has weight $2 < d_*$, which

is impossible; so $|u'| \le |u| - 1$. Also $|\mathrm{supp}(u') \cap \mathrm{supp}(u)| = |u'| - 1 \ge d_* - 1 \ge 2$, so $|u''| = |\mathrm{supp}(u) \setminus \mathrm{supp}(u')| + 1 \le |u| - 1$. By induction on u' (target i, antecedent the restriction of α to $\mathrm{supp}(u') \setminus \{i\} \subseteq \mathrm{supp}(u) \setminus \{j\}$), unit propagation derives the value v_i of x_i determined by α. The assignment $\alpha \cup \{x_i = v_i\}$ is consistent, and its restriction to $\mathrm{supp}(u'') \setminus \{j\} \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$ is an antecedent for u'' and target j; by induction unit propagation derives x_j.

The base case is contained in the induction: a circuit of weight $d_* \ge 3$ is indecomposable, since a decomposition would give, by the two bounds above, $|u'|, |u''| \le d_* - 1$, against the minimality of d_*. Necessity: every usable theory over the base variables contains Φ_{ind} (Proposition A.10). Sufficiency: an explicit theory over the base variables equivalent to $\mathcal{S}$ consists of implicates, so adding its clauses to Φ_{ind} keeps every derivation and causes no conflict on consistent antecedents. □

Corollary A.11 (Counting, and when the given equations suffice)**.** *Let $d_* \ge 3$. (a) Every circuit of weight d_* is indecomposable, so every* UP*-usable theory over the base variables has at least $a_* 2^{d_* - 1}$ clauses. (b) The* core*, the r given equations each written as all of its $2^{|row|-1}$ clauses, is* UP*-usable if and only if every indecomposable circuit is one of the given rows. So a circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one intermediate base quantity, and the core suffices for unit propagation exactly when it already contains every such circuit. (c) One parity law on $m \ge 3$ variables needs exactly its 2^{m-1} clauses over the base variables.*

Proof. (a) As in the base case of the proof of Theorem 1.5. (b) The core is equivalent to $\mathcal{S}$, so by Theorem 1.5 it is usable iff it contains Φ_{ind}. A clause on the support of a circuit u belongs to the core iff some row has that support, that is, as a vector is determined by its support, iff u is a row. (c) $\mathrm{Row}(A) = \{0, 1^m\}$, and 1^m is an indecomposable circuit of weight $m = d_* \ge 3$. □

A derived equation that is not a circuit, for example the sum of two circuits with disjoint supports, never needs to be written.

Proof of Corollary 1.6. Every nonzero $u = yA$ ($y \ne 0$) has $u_c = y \cdot c$ at the column c, which is 1 for exactly half of the 2^r vectors c, none of them 0: so u has weight $2^{r-1} = (m+1)/2$. Equal weights make every nonzero derived equation a circuit, and $d_* = 2^{r-1} \ge 4$, so all m of them are indecomposable by Corollary A.11(a). By Theorem 1.5 the unique minimum is $\Phi_{\mathrm{ind}} = \mathrm{PI}(\mathcal{S})$, with $m 2^{(m-1)/2}$ clauses of width $(m+1)/2$. □

For $r = 3, \ldots, 7$ the minimum is 56, 1,920, 1,015,808, about $1.35 \cdot 10^{11}$ and about $1.2 \cdot 10^{21}$ clauses, against r equations; for $r = 3$ a direct computation (code in the accompanying repository) confirms that $\mathrm{PI}(\mathcal{S})$ answers all 2,352 entailed literals and that deleting any one of its 56 clauses breaks usability.

Remark A.12 (The hypothesis $d_* \ge 3$). The natural extension of Theorem 1.5 and Corollary A.11(a) to $d_* = 2$ fails. The law $x_1 = x_2 = x_3 = x_4$ has $a_* = 6$ derived equations of weight 2, so the bound of Corollary A.11(a) would be 12 clauses, but 4 suffice (a directed cycle; see below); likewise the simplex family with $r = 2$ is the law $x_1 = x_2 = x_3$, with $|\mathrm{PI}| = 6 = a_* 2^{d_* - 1}$ and minimum 3. For every d_*, Φ_{ind} is necessary (Proposition A.10) and nothing else is forced (Proposition 1.4), but for $d_* = 2$ it need not be usable: a direct computation (code in the accompanying repository) finds, for example, a law with $m = 6$, $r = 3$ and $d_* = 2$ for which Φ_{ind} has 4 clauses and the least usable theory over the base variables has 8. For $d_* \le 2$ the difficulty is the choice among alternative clauses, not which clauses are forced.

One case is exact. Call $\mathcal{S}$ an *equivalence law* if $\mathrm{Row}(A)$ is spanned by derived equations of weight 2. Then $i \sim j :\iff \mathbf{e}_i + \mathbf{e}_j \in \mathrm{Row}(A)$ partitions the variables that occur in derived equations into classes $J_1, J_2, \ldots$ of sizes at least 2, and *the least* UP*-usable theory over the base variables has exactly* $|J_1| + |J_2| + \cdots$ *clauses: a directed cycle of implications on each class.*

Proof. Every derived equation is a sum of vectors $\mathbf{e}_i + \mathbf{e}_j$ with $i \sim j$, so its restriction to each class has even weight; hence a derived equation of weight at least 3 contains two variables of one class in its support, the circuits are exactly the weight-2 equations, which lie within classes, and there is no derived equation of weight 1. *Lower bound.* Fix a class J, $i \in J$ and $v \in \{0,1\}$. The antecedent $\{x_i = v\}$ is consistent and entails a literal on each $x_{i'}$, $i' \in J \setminus \{i\}$, so unit propagation derives something in round 1, by a clause whose other literals are falsified by $\{x_i = v\}$: a unit clause, which is not an implicate (no variable is fixed), or a 2-clause containing the literal x_i^{1-v}. A 2-clause implicate has as support the support of a derived equation (Proposition A.7), a pair within one class. So each of the $2|J|$ literals on J has its negation in some 2-clause within J; a 2-clause contains two literals, so at least $|J|$ clauses lie within J, and different classes share no clause. *Upper bound.* Fix $x^0 \in \mathcal{S}$; on $\mathcal{S}$, $x_i^{x_i^0} \leftrightarrow x_{i'}^{x_{i'}^0}$ for $i \sim i'$. For $J = \{i_1, \ldots, i_{|J|}\}$ take the $|J|$ implicates $\neg x_{i_t}^{x_{i_t}^0} \vee x_{i_{t+1}}^{x_{i_{t+1}}^0}$ (indices mod $|J|$). A question (α, ℓ) with ℓ on x_j, $j \in J$, has α assigning some x_i with $i \in J$ (Lemma A.6(iii): the circuits are the pairs); if α makes $x_i^{x_i^0}$ true, propagation runs forward around the cycle to x_j, and if false, backward. No conflict arises. □

A direct computation (exhaustive over subsets of PI; code in the accompanying repository) confirms that the minimum is the size of the class for one class of $2, \ldots, 5$ variables, and is 3 for $x_1 = x_2 \oplus 1 = x_3$, and 5 and 6 for two classes of sizes $(2,3)$ and $(3,3)$.

A.4 Intermediate concepts, systems of laws and depth

Proposition A.13 (The chain encoding)**.** *Let $m \geq 4$. Encode $x_1 \oplus \cdots \oplus x_m = b$ by the constraints $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ for $2 \leq i \leq m-2$, and $y_{m-2} \oplus x_m = b$, each three-variable constraint written as its 4 clauses and the last as its 2 clauses: $4m-6$ clauses and $m-2$ auxiliary variables. The encoding is* propagation complete on all variables*: for every partial assignment α to the base and auxiliary variables, unit propagation derives every literal entailed by the encoding and α, and reaches a conflict if and only if α is inconsistent with the encoding. Without auxiliary variables 2^{m-1} clauses are necessary (Corollary A.11(c)) and sufficient.*

Proof. Each constraint's clause set is the set of all prime implicates of that constraint, so (Lemma A.5(a)) it derives its last variable when the others are assigned, and gives a conflict when all are assigned inconsistently. Consider a fixpoint of unit propagation without conflict. Every constraint is then either fully assigned, and satisfied, or has at least two unassigned variables; call the latter *active*. Each variable occurs in at most two constraints, and consecutive constraints share exactly one auxiliary variable, so the graph on the active constraints, joined when they share an unassigned variable, is a disjoint union of paths. Fix an unassigned variable z and a value for it. Process each component breadth-first from a root (in the component of z, a constraint containing z, with z set first). When a constraint is processed, at most one of its variables is already set: the one it shares with its parent, or z at the root. It has at least two unassigned variables, so at least one of them is still free: give the other free ones arbitrary values and solve for that one. This yields a solution extending the fixpoint with either value of z, so no unassigned variable is entailed. If α were inconsistent and no conflict arose, the same construction (without z) would give a solution extending α; so unit propagation reaches a conflict on every inconsistent α, and by soundness only on those. □

The chain encoding is standard. Every CNF encoding of parity with auxiliary variables has at least $3m$ clauses (Emdin et al., 2022), so the chain is within a factor $4/3$ of the least clause count even among encodings that need not be usable.

Proof of Proposition 1.7. Let $\Phi \in \mathcal{E}_m$ be UP_1-usable and α the antecedent of a question; it assigns base variables only. In round 1 a clause fires only if all its literals but one are falsified; auxiliary literals are unassigned, so a clause containing an auxiliary literal fires only if that literal is its only unfalsified one, and then it derives that auxiliary literal. Hence the base literals derived in round 1 are derived by the clauses of Φ over the base variables, Φ_{base}; no conflict occurs, as $\Phi \cup \alpha$ is satisfiable. Each $C \in \Phi_{\mathrm{base}}$ is an implicate of $\mathcal{S}$, as $\Phi \models C$ and $\operatorname{proj} \Phi = \mathcal{S}$. The proof of Lemma A.5(b), which uses only that the firing clause is an implicate of $\mathcal{S}$, gives $\Phi_{\mathrm{base}} \supseteq \mathrm{PI}(\mathcal{S})$. For one parity law PI consists of the 2^{m-1} clauses of its single circuit; for the simplex family, of $2^{(m-1)/2}$ clauses for each of its m circuits (Corollary 1.6). □

A direct computation (code in the accompanying repository) confirms on twelve random laws that deleting any clause of $\mathrm{PI}(\mathcal{S})$, in all its copies, from $\mathrm{PI}(\mathcal{S})$ together with a library with intermediate concepts destroys UP_1-usability, and that the chain-encoded libraries of the simplex family for $r = 3, 4$ are UP-usable but not UP_1-usable.

Proposition A.14 (Libraries with intermediate concepts)**.** *Let* $\mathrm{Circ}_{\mathrm{aux}}(\mathcal{S})$ *write each circuit* u *of an affine law by the cheaper of its full clause set and its chain encoding with its own auxiliary variables: directly if* $|u| \le 4$*, and by the chain if* $|u| \ge 5$*. It is* UP*-usable, of size* $\sum_{\text{circuits } u} \mathrm{sz}(|u|)$*, where* $\mathrm{sz}(w) := 2^{w-1}$ *for* $w \le 4$ *and* $\mathrm{sz}(w) := 4w - 6$ *for* $w \ge 5$*. The same holds for the library over all* $2^r - 1$ *nonzero derived equations, of size at most* $(2^r - 1)(4m - 6)$ *for* $m \ge 4$*.*

Proof. A question (α, ℓ), ℓ on x_j, has a circuit u with $j \in \mathrm{supp}(u) \subseteq \operatorname{dom} \alpha \cup \{j\}$ (Lemma A.6). The encoding of u is propagation complete (Proposition A.13, or Lemma A.5(a) for the full clause set of the one-equation law), and α fixes all of $\mathrm{supp}(u)$ but j, so unit propagation on that encoding alone derives ℓ; unit propagation is monotone in the clause set. Each auxiliary variable is a partial sum of base variables, so the union is satisfied exactly by the extensions of the points satisfying every circuit, and its projection is $\mathcal{S}$, since the circuits span $\mathrm{Row}(A)$ (proof of Proposition A.7); a consistent α therefore gives no conflict. For the second library, a derived equation of weight w costs at most $\max(8, 4w - 6) \le 4m - 6$ clauses when $m \ge 4$. □

The second form has size $O(m2^r)$, in line with the fixed-parameter tractability in the number of equations shown by Gwynne and Kullmann (2014).

Remark A.15 (The simplex family with intermediate concepts)*.* For the simplex family with $r \ge 4$, $\mathrm{Circ}_{\mathrm{aux}}$ has $m(2m - 4)$ clauses of width at most 3 and $m(m - 3)/2$ auxiliary variables, and is UP-usable: a theory of polynomial size. Writing each circuit by the balanced-tree encoding of Proposition 1.8(b) instead of the chain gives the same size and a $\mathrm{UP}_{2\lceil \log((m+1)/2) \rceil - 1}$-usable theory. So with intermediate concepts and enough rounds the gap to GE is polynomial in written length, and $O(\log m)$ bits under time charging.

Proof. Every circuit has weight $w = 2^{r-1} = (m + 1)/2 \ge 8$, so each is chain-encoded with $4w - 6 = 2m - 4$ clauses and $w - 2 = (m - 3)/2$ auxiliary variables; usability is Proposition A.14. With balanced trees, each question is answered inside the encoding of one circuit, as in the proof of Proposition A.14, within $2\lceil \log w \rceil - 1$ rounds by Proposition 1.8(b), and unit propagation with a fixed number of rounds is monotone in the clause set on satisfiable inputs; the auxiliary variables are again partial sums, so the projection is $\mathcal{S}$. The simplex family is computable from

m in polynomial time, so $Kt(A,b) = O(\log m)$, and Theorem 1.3(b) with $r = \log(m+1)$ gives $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathcal{S})) = O(\log m)$. □

A direct computation (code in the accompanying repository) of the listing lengths (1) gives ratios of about 65, 137, 256 and 461 between this library and the r given equations for $r = 4, 5, 6, 7$: polynomial growth, not $2^{(m-1)/2}$; with chains the library at $r = 4$ needs 7 rounds, with balanced trees 5.

Proposition A.16 (Private intermediate concepts add nothing to a system)**.** *Let $\Phi = \bigcup_t \Phi_t$, where Φ_t encodes a constraint c_t on base variables X_t with auxiliary variables Y_t private to it ($Y_t \cap Y_s = \emptyset$ for $t \neq s$, Y_t occurs only in Φ_t, and $\exists Y_t\, \Phi_t \equiv c_t$), and each Φ_t is propagation complete on $X_t \cup Y_t$. For a partial assignment α to the base variables, let its* arc-consistency closure $\mathrm{AC}(\alpha)$ *be the least set of base literals that contains α and is closed under adding, for every t, every base literal on X_t entailed by c_t and the current literals on X_t; call it* contradictory *if it contains two opposite literals or its literals on some X_t are inconsistent with c_t. Then unit propagation on $\Phi \cup \alpha$ reaches a conflict if and only if $\mathrm{AC}(\alpha)$ is contradictory, and otherwise the base literals at its fixpoint are exactly $\mathrm{AC}(\alpha)$. In particular the given equations written directly and the given equations each chain-encoded derive the same base literals and conflict on the same antecedents.*

Proof. Invariant. At every point of the run, the literals on Y_t are entailed by Φ_t and the literals on X_t: they are derived only by clauses of Φ_t, and each derivation step preserves this. Hence a literal derived by a clause of Φ_t is entailed by Φ_t and the current literals on X_t; if it is a base literal, it is entailed by c_t and the current literals on X_t, as $\exists Y_t\, \Phi_t \equiv c_t$. By induction on rounds, every base literal derived before a conflict lies in $\mathrm{AC}(\alpha)$. A conflict arises either from a clause of some Φ_t with all literals falsified, so that the current literals on $X_t \cup Y_t$ are inconsistent with Φ_t and, by the invariant, those on X_t are inconsistent with c_t; or from two opposite derivations, both from one Φ_t if the variable is auxiliary (with the same conclusion), or both base literals of $\mathrm{AC}(\alpha)$. Either way $\mathrm{AC}(\alpha)$ is contradictory. *Conversely,* suppose the run reaches a conflict-free fixpoint γ' with base part β. No clause of Φ_t fires at γ' and Φ_t is propagation complete, so the restriction of γ' to $X_t \cup Y_t$ is consistent with Φ_t and contains every literal it entails with Φ_t. So $\beta|_{X_t}$ is consistent with c_t, and if $c_t \wedge \beta|_{X_t} \models \ell$ then Φ_t and the literals of γ' on $X_t \cup Y_t$ entail ℓ, so $\ell \in \gamma'$. Thus $\beta \supseteq \alpha$ is closed and not contradictory; hence $\beta \supseteq \mathrm{AC}(\alpha)$ and $\mathrm{AC}(\alpha)$ is not contradictory, and with the first part $\beta = \mathrm{AC}(\alpha)$. The direct encoding of one equation (all its clauses) is propagation complete on its own variables by Lemma A.5(a), and the chain encoding by Proposition A.13. □

So intermediate concepts that each belong to one law cannot help a weak reasoner combine laws; the useful intermediate concepts are shared across laws, such as derived equations and their partial sums. On sparse random systems a direct computation (code in the accompanying repository) finds that unit propagation on the given equations fully answers only a part of the sampled questions, between about 0.62 and 0.89 of them depending on the system, while the circuit library answers all of them.

Lemma A.17 (Usability is arc consistency)**.** *If Φ is* UP*-usable for $\mathcal{S}$, then for every partial assignment α to the base variables that is inconsistent with $\mathcal{S}$, unit propagation on $\Phi \cup \alpha$ reaches a conflict. So* UP*-usability is* arc consistency *of a representation in the sense of Gwynne and Kullmann (2014): unit propagation detects every forced base literal and every inconsistency.*

Proof. Take $\alpha' \subseteq \alpha$ minimal inconsistent and a literal x_j^v of α'. Then $\alpha'' := \alpha' \setminus \{x_j^v\}$ is consistent and entails x_j^{1-v}, so unit propagation on $\Phi \cup \alpha''$ derives it. Unit propagation is monotone up to

conflict: in the run on the larger $\Phi \cup \alpha$, every clause that fired in the smaller run either fires again or meets a conflict. So the larger run derives x_j^{1-v} against $x_j^v \in \alpha$, or conflicts earlier. □

Lemma A.18 (Unit propagation as a monotone circuit)**.** *Let Φ be an explicit theory with $\bar{m}$ variables, written length $\|\Phi\|$ and clause width at most w, and let $\mathcal{I}$ be a set of literals (the inputs). (a) There is a monotone circuit of $O(\bar{m}(\|\Phi\| + \bar{m}))$ fan-in-2 AND and OR gates, with one input per literal of $\mathcal{I}$, that outputs, for every literal L, whether unit propagation on $\Phi \cup \alpha$ derives L, on every input $\alpha \subseteq \mathcal{I}$ for which $\Phi \cup \alpha$ is satisfiable. (b) For every k there is a monotone circuit of unbounded fan-in AND and OR gates, of depth $2k$, with at most $k(\|\Phi\| + 2\bar{m})$ gates and at most $k(w\|\Phi\| + 2\bar{m})$ wires, with the same inputs, that outputs whether UP_k derives L, on the same inputs.*

The direction "small usable theory gives small monotone circuit" of the equivalence of Bessiere et al. (2009) is (a).

Proof. Define $o_0(L)$ as the input bit if $L \in \mathcal{I}$ and the constant 0 otherwise, and $o_{t+1}(L) := o_t(L) \vee \bigvee_{C \ni L} \bigwedge_{L' \in C,\, L' \neq L} o_t(\neg L')$, where the empty conjunction, for a unit clause $C = (L)$, is the constant 1. On a satisfiable $\Phi \cup \alpha$ unit propagation never conflicts, so a clause whose other literals are all falsified fires (its last literal is unassigned or already true), exactly as in the formula; by induction $o_t(L)$ says whether L is known after t rounds. (a) Prefix and suffix products give the $|C|$ conjunctions of a clause with $3|C|$ fan-in-2 gates, so a round costs $O(\|\Phi\| + \bar{m})$ gates; each round before the fixpoint assigns at least one new variable and no variable is assigned twice, so $\bar{m}$ rounds suffice. (b) Per round, one AND gate for each pair (clause, literal of it), of fan-in $|C| - 1 \leq w - 1$, and one OR gate for each literal, of fan-in one plus the number of clauses containing it: at most $\|\Phi\| + 2\bar{m}$ gates and $w\|\Phi\| + 2\bar{m}$ wires per round. AND layers and OR layers alternate (an OR gate feeding an OR gate of the next round is merged into it), so the depth is $2k$. Constants are removed by simplification without increasing size or depth. □

Proof of Theorem 1.9. The law. $|\Lambda_q| = 2\binom{q}{3} + 2q$, so $m = 2\binom{q}{3} + 2q + 1 = \Theta(q^3)$. The vectors of Λ_q span $\mathrm{GF}(2)^{q+1}$ (already $(\mathbf{e}_v, 0)$ and $(\mathbf{e}_v, 1)$, $v \leq q$, do), so $\dim \mathcal{H}_q = q + 1$ and $\mathcal{H}_q$ is an affine law of rank $r = m - q - 1 \geq 1$. The m equations $x_{(u,a)} = u \cdot y + a\, z_0$, $x_\star = z_0$, with $z = (y, z_0)$ as auxiliary variables, have projection $\mathcal{H}_q$, and GE returns exactly the entailed base literals, so they are GE-usable. *The lower bound.* For $S \subseteq \Lambda_q$ the antecedent $\alpha_S := \{\neg x_v : v \in S\}$ is consistent, as $0 \in \mathcal{H}_q$. On $\mathcal{H}_q$, $x_\star$ is determined by x_S iff $\mathbf{e}_\star \in \mathrm{span}(S)$ (if not, some z has $z \cdot v = 0$ for $v \in S$ and $z \cdot \mathbf{e}_\star = 1$, and z and 0 give two points that agree on S and differ at $x_\star$; if so, $x_\star$ is a sum of the x_v, $v \in S$), iff the XOR system $\{u \cdot y = a : (u, a) \in S\}$ over $y \in \mathrm{GF}(2)^q$ is unsatisfiable (a subset of the equations sums to $0 = 1$, that is, a subset of S sums to $\mathbf{e}_\star$). If $x_\star$ is determined, its value is that at the point 0, so usability gives "unit propagation on $\Phi \cup \alpha_S$ derives $\neg x_\star$"; if not, soundness ($\mathrm{proj}\, \Phi = \mathcal{H}_q$) excludes any literal on $x_\star$. So the output for $\neg x_\star$ of the circuit of Lemma A.18(a), with inputs $\mathcal{I} = \{\neg x_v : v \in \Lambda_q\}$, computes the monotone function $\mathrm{Unsat}_q(S) :=$ [the system S is unsatisfiable], with $O((\|\Phi\| + m)^2)$ gates (auxiliary variables that do not occur in Φ are dropped, so $\bar{m} \leq \|\Phi\| + m$). Göös et al. (2019) show that a monotone variant of XOR-SAT has exponential monotone circuit complexity. In the form we use, their function $\mathrm{3XorSat}_q$ has one input for each ordered triple $(v_1, v_2, v_3) \in [q]^3$ and each constant $a \in \{0, 1\}$ ($2q^3$ inputs), accepts iff the selected constraints $y_{v_1} \oplus y_{v_2} \oplus y_{v_3} = a$ are unsatisfiable, and requires monotone circuits of size 2^{q^ε} for some $\varepsilon > 0$. Each input (v_1, v_2, v_3, a) is the constraint $u \cdot y = a$ with $u = \mathbf{e}_{v_1} + \mathbf{e}_{v_2} + \mathbf{e}_{v_3}$, of weight 1 (if two indices coincide) or 3, so $(u, a) \in \Lambda_q$; hence $\mathrm{3XorSat}_q(Z) = \mathrm{Unsat}_q(\mathrm{pr}(Z))$, where $\mathrm{pr}(Z)_v$ is the OR of the inputs that map to v. A monotone circuit for Unsat_q together with at most $2q^3$ OR gates is a monotone circuit for $\mathrm{3XorSat}_q$. (If the inputs are only the triples of distinct indices,

$3\mathrm{XorSat}_q$ is Unsat_q with the weight-1 inputs set to 0, and a circuit for Unsat_q again gives one for it.) Hence $O((\|\Phi\|+m)^2)+2q^3 \geq 2^{q^\varepsilon}$, so $\|\Phi\| \geq 2^{q^\varepsilon/2-O(1)} - m$ for $q \geq q_0$, and $q^\varepsilon = \Theta(m^{\varepsilon/3})$. □

The weight-1 functionals matter: with weight-3 functionals only, the function computed would be the restriction of $3\mathrm{XorSat}_q$ that sets the inputs with repeated indices to 0, and a monotone lower bound does not pass to a restriction. A direct computation (code in the accompanying repository) confirms for $q = 3, 4$ that "$x_\star$ determined", "system unsatisfiable" and "$\mathbf{e}_\star \in \mathrm{span}(S)$" agree on all $2^{|\Lambda_q|}$ sets S, and that unit propagation on $\mathrm{PI}(\mathcal{H}_3)$ derives $\neg x_\star$ from α_S exactly when the system is unsatisfiable.

Remark A.19 (XOR systems without auxiliary-variable restrictions). Gwynne and Kullmann (2014) show that systems of XOR constraints have no polynomial-size arc-consistent representation in general, even with auxiliary variables, and that the standard translation is propagation complete for one equation and fails for two, the counterpart with auxiliary variables of Corollary A.11(b). By Lemma A.17 their notion is our UP-usability, so their superpolynomial bound applies to usable theories; Theorem 1.9 gives an explicit family with an exponential bound.

Proof of Proposition 1.8. For one parity law every question has $|\mathrm{dom}\,\alpha| = m-1$, as the only circuit is 1^m. (a) Write $C_1 = (x_1, x_2, y_1)$, $C_i = (y_{i-1}, x_{i+1}, y_i)$ for $2 \leq i \leq m-2$, and $C_{m-1} = (y_{m-2}, x_m)$ for the constraints. Each constraint's clauses mention only its own variables and fire only when all but one of them are assigned. *Upper bound:* let x_j be the unknown base variable and $C_{j'}$ its constraint ($j' = \max(1, j-1)$). The constraints to the left of $C_{j'}$ resolve from the left end (C_1 derives y_1 in round 1, C_2 derives y_2 in round 2, and so on), and those to the right from the right end (C_{m-1} derives y_{m-2} in round 1, and so on); $C_{j'}$ then has all its variables but x_j assigned by round $\max(j'-1, m-1-j')$, and derives x_j one round later, by round $m-1$. *Lower bound:* for the question with x_1 unknown, after round $t \leq m-2$ exactly the auxiliary variables $y_{m-2}, \ldots, y_{m-1-t}$ are assigned (induction on t: C_1 has the two unknowns x_1, y_1, and each C_i, $2 \leq i \leq m-2$, the two unknowns y_{i-1}, y_i, until the boundary reaches it, so in each round exactly one constraint has a single unknown); so x_1 is derived in round $m-1$ and not before. (b) Let $H := \lceil \log m \rceil$ be the height of the tree. Each constraint's clause set is its set of prime implicates, so it derives its third variable in the round after the other two are known (Lemma A.5(a)). Let x_j be the unknown leaf and $v_0, \ldots, v_s = x_j$ the path from the root, $s \leq H$; for $1 \leq i \leq s$ let w_i be the sibling of v_i. Every node v not on the path has only known leaves below it, and by induction on its height $\mathrm{ht}(v)$ its value is derived by round $\mathrm{ht}(v)$; w_i is at depth i, so $\mathrm{ht}(w_i) \leq H - i$. The root constraint gives y_{v_1} by round $\mathrm{ht}(w_1)+1 \leq H$. If y_{v_i} is known by round $H+i-1$, the constraint at v_i gives $y_{v_{i+1}}$ by round $\max(H+i-1, H-i-1)+1 = H+i$. So x_j is derived by round $H+s-1 \leq 2H-1$. No conflict arises, as the antecedent is consistent. Counting: there are $m-1$ internal nodes, $m-2$ of them non-root with 4 clauses each, and the root with 2. (c) Let Φ be UP_k-usable, and drop auxiliary variables that do not occur, so that $\bar{m} \leq m + \|\Phi\|$. Apply Lemma A.18(b) with inputs the literals on $x_2, \ldots, x_m$ (for a general affine law: on $\mathrm{supp}(u) \setminus \{j\}$ for a circuit u of weight w and some $j \in \mathrm{supp}(u)$). Every assignment to these variables is consistent (no nonzero derived equation is supported among them) and determines x_1 (resp. x_j) as a constant plus their parity (Lemma A.6); so by usability and soundness the output gate for the literal x_1 computes, on the inputs that make exactly one literal of each variable true, the parity of $m-1$ bits or its negation. This is a circuit of depth $2k$ with unbounded fan-in AND and OR gates over the literals of $m-1$ variables, with at most $k(\|\Phi\|+2\bar{m}) \leq k(3\|\Phi\|+2m)$ gates. By Håstad's theorem (Håstad, 1986), in its standard form, every such circuit of depth $d \geq 2$ computing the parity of n variables has at least $2^{c_{\mathrm{H}} n^{1/(d-1)}}$ gates, for an absolute constant $c_{\mathrm{H}} > 0$ and n large enough given d. With $n = m-1$ (resp. $w-1$) and $d = 2k$ the claim follows. (For $k = 1$, Proposition 1.7 gives the exact

value 2^{m-1}.) (d) By (c), for fixed k the written length is $2^{\Omega(m^{1/(2k-1)})}$, superpolynomial in m; and comparing a polynomial size $2^{O(\log m)}$ with $2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$ gives $(m-1)^{1/(2k-1)} = O(\log m)$, that is, $k = \Omega(\log m/\log\log m)$. This comparison applies (c) with k growing with m, that is, Håstad's bound for depths up to this order. By (b), $2\lceil\log m\rceil - 1$ rounds suffice for $4m-6$ clauses. □

A direct computation over all $m2^{m-1}$ questions, $m = 3, \ldots, 10$ (code in the accompanying repository), confirms that the chain needs exactly $m-1$ rounds for $m \geq 4$ and that the balanced tree needs $1, 3, 4, 5, 5, 5, 6, 7$ rounds for $m = 3, \ldots, 10$, within the bound $2\lceil\log m\rceil - 1$. Part (c) is a direct application of Håstad's theorem and very probably folklore.

Proof of Corollary 1.10. First column. One parity law has the single circuit 1^m, indecomposable, so every UP-usable theory over the base variables, and a fortiori every UP_k-usable one, contains its 2^{m-1} clauses (Proposition A.10), and these clauses are UP_1-usable (Corollary A.11(c), Proposition A.7). *Row* UP_1. Proposition 1.7, with $\mathrm{PI}(\mathcal{S})$ itself UP_1-usable. *Row* UP_k, *k fixed.* Proposition 1.8(c), for parity with $w = m$ and for the simplex family with $w = (m+1)/2$. *Row* UP_k, *$k \geq 2\lceil\log m\rceil - 1$, and* UP. Proposition 1.8(b) and Remark A.15 (for $r = 3$, $\mathrm{PI}(\mathcal{S})$ has $56 \leq m(2m-4)$ clauses); more rounds never hurt. *Last column.* Theorem 1.9, for every number of rounds. *Row* GE. A linear theory with e equations in $m + m_{\mathrm{aux}}$ variables has a solution set of dimension at least $m + m_{\mathrm{aux}} - e$, whose projection has dimension at least $m - e$; so a theory for a law of rank r has at least r equations, and the given equations (one for parity, $r = \log(m+1)$ for the simplex family) or the m equations of Theorem 1.9 are usable. □

A.5 Directionality

A *directed rule* is $P \Rightarrow \ell$, with P a set of literals (the premise) and ℓ a literal on a variable that does not occur in P; a "rule" whose conclusion's variable occurs in its premise corresponds to a tautological clause and is excluded, as tautological clauses are excluded from explicit theories. Its written length is $|P| + 1$. *Forward chaining* FC_k applies, in each of k parallel rounds, every rule whose premise is satisfied by the current assignment; a conclusion whose opposite is assigned, or two opposite conclusions, give a conflict; $\mathrm{FC} := \mathrm{FC}_\infty$. A rule over the base variables is *sound* for $\mathcal{S}$ if P is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge P \models \ell$. The projection of a rule set, possibly with auxiliary variables, is that of its clauses $\neg P \vee \ell$; a rule set is FC_k*-usable* for $\mathcal{S}$ if its projection is $\mathcal{S}$ and FC_k derives every entailed base literal from every consistent antecedent. A *rule base* is a set of sound rules over the base variables.

Proposition A.20 (Directed rules against clauses)**.** *(a) For a clause C let* $\mathrm{rules}(C) := \{\neg(C \setminus \{L\}) \Rightarrow L : L \in C\}$*, where $\neg(C \setminus \{L\})$ is the set of negations of the other literals. For every theory Φ, every k and every α,* FC_k *with* $\mathrm{rules}(\Phi)$ *and* UP_k *with Φ derive the same literals and conflict together. Conversely a rule $P \Rightarrow \ell$ gives the clause $\neg P \vee \ell$, and unit propagation with the clause derives at least what forward chaining with the rule derives. Hence, for every law and every round budget k, with or without auxiliary variables, the smallest usable rule set has at least as many rules as the smallest usable clausal theory has clauses, and at most w times as many, w the largest clause width of a smallest usable clausal theory; in written length the factor is likewise at most w. (b)* One parity law on $m \geq 2$ variables, no auxiliary variables. *A sound rule concluding a literal on x_j has all other $m-1$ variables in its premise. So there are exactly $m2^{m-1}$ sound rules, each question needs its own rule, and the only usable rule base is the set of all of them: $m2^{m-1}$ rules of length m, against the unique minimum of 2^{m-1} clauses*

of length m; a factor m, the number of query directions, in count and in written length. (c) With auxiliary variables, for every law. *Splitting each clause $(l_1 \vee \cdots \vee l_w)$ with $w \geq 4$ into $(l_1 \vee l_2 \vee z_1), (\neg z_1 \vee l_3 \vee z_2), \ldots, (\neg z_{w-3} \vee l_{w-1} \vee l_w)$ with fresh variables $z_1, \ldots, z_{w-3}$ preserves the projection and* UP*-usability. So every law has an* FC*-usable rule set of written length at most* 9 *times the least written length of a* UP*-usable clausal theory with auxiliary variables. For the chain encoding of parity: $4m - 6$ clauses against $3(4m - 6) - 2$ rules. (d)* One round, any vocabulary. *For one parity law on $m \geq 2$ variables, every FC_1-usable set of sound rules, with any auxiliary variables, contains all $m2^{m-1}$ sound rules over the base variables, while every UP_1-usable theory with any auxiliary variables must contain the 2^{m-1} clauses of $\mathrm{PI}(\mathcal{S})$ (Proposition 1.7).*

Proof. (a) Induction on rounds: a clause C fires in round t deriving L exactly when the rule $\neg(C \setminus \{L\}) \Rightarrow L$ has its premise satisfied after round $t - 1$ and L is not yet assigned; forward chaining signals a conflict when a conclusion's opposite is assigned, as unit propagation does when a clause is falsified. A rule $P \Rightarrow \ell$ whose premise is satisfied leaves its clause with all literals but ℓ falsified. The bounds follow by translating a smallest usable theory in either direction: rules(Φ) has at most $w|\Phi|$ rules and written length $\sum_{C \in \Phi} |C|^2 \leq w\|\Phi\|$, and the clauses of a rule set are at most as many as its rules and are usable for the same round budget. (b) If a sound rule $P \Rightarrow x_j^v$ had a premise missing some x_i, $i \neq j$, then by Lemma A.6 (the only circuit is 1^m) x_j would not be determined by the consistent premise P, and the rule would be unsound. So each sound rule is $\beta|_{[m]\setminus\{j\}} \Rightarrow x_j^v$ with v the forced value, one for each j and each of the 2^{m-1} assignments β to the other variables, and the question with that antecedent is answered only by that rule. The minimum of 2^{m-1} clauses is Corollary A.11(c) for $m \geq 3$, and for $m = 2$ the equivalence-law case of Remark A.12 with one class of two variables. (c) The pieces, with the z's quantified out, are equivalent to the clause. If all literals of the clause but l_t are falsified, propagation from the left derives $z_1, \ldots, z_{t-2}$, from the right $\neg z_{w-3}, \ldots, \neg z_{t-1}$, and the piece containing l_t fires (the end cases are the same with one side empty). Every derivation of the original run is reproduced, in more rounds, and soundness is unchanged. A clause of length $w \geq 4$ becomes $w - 2$ clauses of length 3, of total length at most $3w$, and each 3-clause becomes 3 rules of length 3; clauses of width at most 3 become rules of total length at most 3 times their own. The chain encoding has $4m - 8$ clauses of width 3 and 2 of width 2, so rules gives $3(4m - 8) + 4 = 3(4m - 6) - 2$ rules. (d) From an antecedent α on base variables, a rule fires in round 1 only if its premise is satisfied by α, so its premise consists of base literals of α. A rule answering the question with antecedent $\beta|_{[m]\setminus\{j\}}$ concludes a literal on x_j, so it is a sound rule over the base variables, and by (b) its premise is that whole antecedent. □

A direct computation (code in the accompanying repository) confirms for $m = 2, \ldots, 7$ that the sound rules of one parity law number exactly $m2^{m-1}$, all with premises of size $m - 1$, and that for $m \leq 5$ deleting any one of them breaks usability; on random theories, forward chaining with rules(Φ) and unit propagation with Φ agree in every run.

Theorem A.21 (Directionality is real but cheap)**.** *(1) For one parity law on $m \geq 2$ variables, the only* FC*-usable rule base is the set of all $m2^{m-1}$ sound rules, of written length m each, while the least* UP*-usable theory over the base variables has 2^{m-1} clauses of length m: a factor m, the number of query directions. With any auxiliary variables, every FC_1-usable set of sound rules contains these $m2^{m-1}$ rules, while every UP_1-usable theory must contain 2^{m-1} clauses. (2) For every law and round budget k, with or without auxiliary variables, the least FC_k-usable rule set has at least as many rules as the least UP_k-usable theory has clauses and at most w times as many, w the largest clause width of a smallest such theory; in written length the factor is likewise*

at most w. With auxiliary variables and unbounded rounds, every law has an FC*-usable rule set of written length at most* 9 *times the least written length of a* UP*-usable theory (at most* 3 *times for the chain encoding of parity). A program with internal variables answers all m directions of parity with $O(m)$ operations. (3) $K_A^{\mathrm{FC}_k} = K_A^{\mathrm{UP}_1} \pm c$ on every conditional data set.*

Proof. (1) is Proposition A.20(b), (d). (2) is Proposition A.20(a), (c); the chain encoding has width at most 3, so its rules have written length at most 3 times its own; and a program answers the question with x_j unknown by outputting b plus the sum of the given bits. (3) A fixed program maps an FC_k-compatible rule set R to its clauses and then to PI of their projection, which is UP_1-compatible with the same data by Lemma A.5(a), since forward chaining derives only literals entailed by the clauses; conversely it maps a UP_1-compatible Φ to rules(Φ), which is FC_1-compatible by Proposition A.20(a), hence FC_k-compatible for every $k \geq 1$. □

Remark A.22 (Programs, rule bases and time bounds)*.* A program may consist of a generic solver applied to a declarative law, at $O(1)$ extra length, so "programs lose" cannot be the statement; Theorem 1.2 is the precise form for reasoners: under program length the weakest reasoner together with a compiler is as good as any. The separations above concern what a *fixed* weak reasoner needs written. The translation of Proposition A.20(a) is round for round, so the directionality factor is the same under every round budget. A per-question time bound below that of elimination ($O(r^2 m)$ bit operations for GE on r equations) does not separate directed rules from clauses either: PI($\mathcal{S}$) and its rule translation both answer every question of the simplex family in one round. What a time bound changes is the comparison between reasoners: a reasoner that may not run elimination must be handed a lemma library, which over the base variables is exponentially larger (Appendix A.3); if its time is charged at least linearly in the size of the theory it reads, it cannot even read that library.

Question A.23. In a cell-probe-type model of a reasoner, is there a theory of size polynomial in m that answers every question of the simplex family in time below that of elimination?

A.6 Where the know-how lives

Proposition A.24 (Three homes of know-how)**.** *For an affine law $\mathcal{S} = \{x : Ax = b\}$, the know-how to answer every question of $\mathcal{Q}(\mathcal{S})$ can be placed in any of the following. (a)* In the reasoner: *the r equations with* GE*, of listing length $O(rm \log m)$ bits. (b)* In written lemmas: *for* UP *over explicit theories, exactly $\Phi_{\mathrm{ind}}(\mathcal{S})$ over the base variables when $d_* \geq 3$ (Theorem 1.5; exponential for the simplex family), and a library such as $\mathrm{Circ}_{\mathrm{aux}}$ with intermediate concepts (Proposition A.14; polynomial for the simplex family by Remark A.15, exponential for some systems whatever the intermediate concepts by Theorem 1.9, and for one parity law at any fixed depth by Proposition 1.8(c)). (c)* In a computation that the cost model does not charge: *either a membership test with free search for derivations: for a suitable polynomial q, the axiom set* PI($\mathcal{S}$) *has a decider of length $K^q(A, b) + c$ that runs in time polynomial in m and in the length of its input, and for every $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$ the sentence $\bigwedge \alpha \to \ell$ follows by propositional reasoning from a single member of* PI($\mathcal{S}$)*; or a generator of the explicit theory under program length (Theorem 1.2: the compiler $\Phi' \mapsto \mathrm{PI}(\mathrm{proj}\, \Phi')$ costs $O(1)$ bits). Here $K^q(A, b) \leq rm + r + O(\log m)$, and $K^q(A, b) = O(\log m)$ for one parity law and for the simplex family.*

Proof. (a) and (b) collect the results cited. (c) A clause $C_{Z,\beta}$ is in PI($\mathcal{S}$) iff the derived equations supported in Z are exactly 0 and the all-ones vector 1_Z on Z, and β violates 1_Z (Proposition A.7; a circuit is the unique nonzero derived equation supported in its support). The derived equations

supported in Z are the yA with y in the kernel of the columns of A outside Z, computable by Gaussian elimination in time polynomial in m once (A, b) is available, and a program of length $K^q(A, b)$ produces (A, b) within $q(|(A, b)|)$ steps. A question (α, ℓ) has a prime implicate $C = \neg\alpha' \vee \ell$ with $\alpha' \subseteq \alpha$ (Lemma A.5(a)), and $\bigwedge \alpha \to \ell$ follows from the single instance C by propositional reasoning. Printing the literal matrix and vector, with r and m coded self-delimitingly, takes polynomial time and gives $K^q(A, b) \leq rm + r + O(\log m)$; one parity law and the simplex family are computed from m in polynomial time. □

A direct computation (code in the accompanying repository) confirms that the decider agrees with prime-implicate membership computed by brute force on 9,852 clauses over 30 random laws.

Definition A.25 (Charged deduction with conditional data)**.** We use the readings of charged deduction of **??** (derivation size; every instance short, (a1), or every nonlogical instance short, (a2); total material, (b), or nonlogical material, (b2)) at polynomial budgets with the degrees charged, as in Definition E.18 of Hänni and Claude (Anthropic) (2026). A hypothesis is a triple (p, i, j); $A_{p,i}$ is the set of sentences χ that p accepts within $(|\chi| + 2)^i$ steps; the budget for a datum φ is $(|\varphi| + 2)^j$; and the weight is $2^{-|\mathrm{El}(p)| - 2\lceil\log(i+1)\rceil - 2\lceil\log(j+1)\rceil - 2}$, with $\mathrm{El}(p)$ a self-delimiting code of p of length $|p| + O(\log|p|)$. The *conditional* version: (p, i, j) is compatible with a conditional data set D if for each $(\alpha, \ell) \in D$ the set $B \cup A_{p,i} \cup \{\alpha\}$ is consistent (this is not charged) and the sentence $\bigwedge \alpha \to \ell$ has a derivation in the calculus K of that report from $B \cup A_{p,i}$ within the budget under the reading. In our setting $\mathcal{L}_m$ is finite, $B = \emptyset$ is decidable in polynomial time, and **??** holds vacuously, as the axioms are sentences of $\mathcal{L}_m$, which contain no parameters.

The readings are defined for two-valued data; question data reach them only through the two-valued encoding, which is degenerate (Proposition A.3), or through this definition.

Theorem A.26 (Charged deduction does not charge for the lemma library)**.** *Let $\mathcal{S}$ be an affine law and take any reading of charged deduction in Definition A.25, at polynomial budgets with the degrees charged. (i) If each derivation of Proposition A.24(c) has size at most $(|\bigwedge \alpha \to \ell| + 2)^{j_0}$, the decider of $\mathrm{PI}(\mathcal{S})$ with derivation degree j_0 is a hypothesis compatible with $\mathcal{Q}(\mathcal{S})$ of weight at least $2^{-K^q(A,b) - O(\log m) - 2\lceil\log(j_0+1)\rceil}$. (ii) Every hypothesis compatible with $\mathcal{Q}(\mathcal{S})$ has weight at most $2^{-K(\mathcal{S})+c}$.*

Proof. (i) By Proposition A.24(c) the decider runs in time $\mathrm{poly}(m) \cdot \mathrm{poly}(|\chi|) \leq (|\chi| + 2)^i$ with $i = O(\log m)$, as $|\chi| + 2 \geq 3$, so the degree charge is $O(\log\log m)$; it has length $K^q(A, b) + c$, so $|\mathrm{El}(p)| = K^q(A, b) + O(\log m)$. $\mathrm{PI}(\mathcal{S}) \cup \alpha$ is consistent for every antecedent of $\mathcal{Q}(\mathcal{S})$, and by hypothesis each datum has a derivation from a single member of $\mathrm{PI}(\mathcal{S})$ of size at most $(|\bigwedge \alpha \to \ell| + 2)^{j_0}$, which bounds its instances and its material as well, so it is within the budget under every reading. (ii) Let (p, i, j) be compatible with $\mathcal{Q}(\mathcal{S})$ and $\Gamma := A_{p,i}$. *Claim:* $x \notin \mathcal{S}$ iff there are $I \subsetneq [m]$ and $j' \notin I$ such that $\bigwedge x|_I \to \ell$ has a derivation from Γ within the budget, where ℓ is the literal on $x_{j'}$ false at x. If $x \notin \mathcal{S}$, the proof of Lemma A.2 gives a question $(x|_I, \ell) \in \mathcal{Q}(\mathcal{S})$ with ℓ false at x, and compatibility gives the derivation. If $x \in \mathcal{S}$, then x has a determined coordinate j'' (Lemma A.2), and the datum $(x|_{[m]\setminus\{j''\}}, \ell'')$, with ℓ'' the literal on $x_{j''}$ true at x, makes $\Gamma \cup \{\hat{x}\}$ consistent; a derivation of $\bigwedge x|_I \to \ell$ with ℓ false at x would refute $\Gamma \cup \{\hat{x}\}$. The condition is decidable from (p, i, j): there are finitely many derivations within the budget, and membership in Γ is decided by running p with its clock. So $\mathcal{S}$ is computable from (p, i, j), and $K(\mathcal{S}) \leq |\mathrm{El}(p)| + 2\lceil\log(i + 1)\rceil + 2\lceil\log(j + 1)\rceil + c$, which is the claim. □

Remark A.27 (Derivation sizes, and the two-valued encoding under charged deduction). We do not count the nodes of the derivations below. (1) In the calculus K (connectives $\neg$ and $\to$, with

$\vee$ and $\wedge$ defined), the tautology $(\neg\alpha' \vee \ell) \to (\bigwedge \alpha \to \ell)$ has a derivation of size polynomial in $|\alpha| + |\ell| = O(m)$, assembled from the standard derivations of the deduction theorem, $\wedge$-elimination and the double-negation laws, used $O(|\alpha|)$ times; one modus ponens with the axiom $C = \neg\alpha' \vee \ell$ finishes. So the hypothesis of Theorem A.26(i) holds with $j_0 = O(1)$, the decider has weight at least $2^{-K^q(A,b)-O(\log m)}$, and by Theorem A.26(ii) these readings give the full lemma library, whose explicit form has $m2^{(m-1)/2}$ clauses for the simplex family, a cost within $K^q(A,b) - K(\mathcal{S}) + O(\log m)$ bits of the least possible. (2) Under the two-valued encoding these readings are degenerate as SAI is: the point theory T_{x^0} of Proposition A.3 has a polynomial-time decider and is consistent, and the case distinction in the proof of Proposition A.3 gives each $\bigwedge \alpha \to \ell$ from one literal of T_{x^0} by propositional steps of polynomial size.

Charged deduction thus measures the length of a membership test, not the length of what a reader must be handed; the membership test for the full library runs Gaussian elimination, so the know-how lives in the decider. On literal sequences, under **??**, at polynomial budgets with the degrees charged, charged-deduction SAI corresponds to certificate SOI (**????**), and the nondeterminism is the free search for derivations.

A.7 Learning know-how pattern by pattern

Let $m \geq 3$ and $\mathrm{Par}_m := \mathrm{Par}_m^0$ (the constant 1 is symmetric). Every question of $\mathcal{Q}(\mathrm{Par}_m)$ has an antecedent on $m-1$ variables; write it as (j,ζ) with $\zeta \in \mathrm{Par}_m$, its *positive point*, antecedent $\zeta|_{[m]\setminus\{j\}}$ and answer $x_j^{\zeta_j}$. Its *pattern* is the violating point $\mathrm{pat}(j,\zeta) := \zeta \oplus \mathbf{e}_j \in \mathcal{P}_m := \{0,1\}^m \setminus \mathrm{Par}_m$. So $|\mathcal{Q}(\mathrm{Par}_m)| = m2^{m-1}$ and $|\mathcal{P}_m| = 2^{m-1}$, and every pattern and every positive point carries exactly m questions. For a conditional data set $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, $\mathrm{pat}(D)$ is the set of *seen* patterns. For $\beta \in \{0,1\}^m$, C_β is the full-width clause falsified exactly by β; in the listing code it costs $\mathrm{len}_m := \ell_{\mathbb{N}}(m) + m(\lceil \log m \rceil + 1)$ bits. A theory *fails* on a question if it errs, abstains, or is inconsistent with the antecedent.

Lemma A.28 (Answers of a theory over the base variables)**.** *For an explicit theory Φ over the base variables and a question (j,ζ) the following are equivalent: (i) Φ is* UP*-compatible with (j,ζ); (ii) Φ answers it correctly (unit propagation derives exactly $x_j^{\zeta_j}$, without conflict); (iii) ζ satisfies Φ, and some clause of Φ with a literal on x_j is falsified by* $\mathrm{pat}(j,\zeta)$.

Proof. Only x_j is unassigned, so unit propagation either finds a clause falsified by the antecedent (a conflict), or derives values of x_j from clauses whose other literals are falsified by it. A clause with a literal on x_j and all other literals falsified by the antecedent derives $x_j^{\zeta_j}$ iff it is falsified by $\mathrm{pat}(j,\zeta)$, and derives the opposite iff it is falsified by ζ; a clause without x_j that is falsified by the antecedent is falsified by ζ. So (iii) gives a derivation of the answer and no conflict, that is (ii), and ζ witnesses satisfiability, so (i) follows. If (i), the satisfying completion of the antecedent is ζ, since the clause that derives $x_j^{\zeta_j}$ excludes $\mathrm{pat}(j,\zeta)$; this gives (iii). (ii)$\Rightarrow$(iii): a correct derivation needs a clause falsified by $\mathrm{pat}(j,\zeta)$, and a clause falsified by ζ would cause a conflict or a wrong derivation. □

Theorem A.29 (A weak reasoner learns its know-how pattern by pattern)**.** *(a)* Full correctness. *An explicit theory over the base variables answers every question of $\mathcal{Q}(\mathrm{Par}_m)$ correctly iff it is* $\mathrm{PI}(\mathrm{Par}_m) = \{C_\beta : \beta \in \mathcal{P}_m\}$. *(b)* Minimal theories, any order. *Let $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$ and let Φ be an inclusion-minimal explicit theory over the base variables that is* UP*-compatible with D (for instance any ℓ_{lst}-shortest one, Lemma A.1). Then for every unseen pattern $\beta \notin \mathrm{pat}(D)$, Φ fails on some question of pattern β. Hence Φ fails on at least $|\mathcal{P}_m \setminus \mathrm{pat}(D)|$ questions, and*

a shortest compatible theory is fully correct only after all 2^{m-1} patterns have been seen. (c) Random order. *If the questions are drawn independently and uniformly from $\mathcal{Q}(\mathrm{Par}_m)$, the patterns are independent and uniform on $\mathcal{P}_m$; after n questions the expected number of unseen patterns is $2^{m-1}(1-2^{1-m})^n$, and the expected time until all have been seen is $2^{m-1}\sum_{i=1}^{2^{m-1}} 1/i = 2^{m-1}((m-1)\ln 2 + 0.5772\ldots) + \frac{1}{2} + O(2^{-m})$, where $0.5772\ldots$ is Euler's constant. (d)* Bayes, block-complete data (semimeasure convention). *If D contains all m questions of each seen pattern, every* UP*-compatible explicit theory over the base variables contains $\{C_\beta : \beta \in \mathrm{pat}(D)\}$, and $-\log W_{\mathrm{lst}}^{\mathrm{UP,base}}(D) \geq |\mathrm{pat}(D)|\,\mathrm{len}_m$. So the cumulative loss is at least $|\mathrm{pat}(D)|\,\mathrm{len}_m - c$ bits: about $m \log m$ bits per pattern, and $2^{m-1}\mathrm{len}_m$ in all. (e)* With the reasoner or with intermediate concepts. *For every $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, $-\log W_{\mathrm{lst}}^{\mathrm{GE}}(D) \leq \ell_{\mathrm{lst}}$(the one equation) $= m\lceil \log m \rceil + O(\log m)$, and $-\log W_{\mathrm{lst}}^{\mathrm{UP}}(D) \leq \ell_{\mathrm{lst}}$(the chain encoding) $= O(m \log m)$ (Proposition A.13). So these inducers pay $O(m \log m)$ bits in total on every question sequence. (f)* Other measures. *Under program length there is no gap: $K_A^{\mathrm{UP,base}}(\mathcal{Q}(\mathrm{Par}_m)) = K(m) \pm c$. Under time charging, $Kt_A^{\mathrm{UP,base}}(\mathcal{Q}(\mathrm{Par}_m)) \geq \log(m2^{m-1}) = m - 1 + \log m$, while $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathrm{Par}_m))$ and $Kt_A^{\mathrm{GE}}(\mathcal{Q}(\mathrm{Par}_m))$ are $O(\log m)$.*

Proof. (a) By Lemma A.28(iii) for all questions: every point of Par_m satisfies Φ, so every clause is an implicate, of width m (Proposition A.8, $d_* = m$), that is, some C_β with $\beta \in \mathcal{P}_m$; and every pattern β needs a clause falsified by it, which among full-width clauses is C_β itself. (b) C_β is falsified only by β, so by Lemma A.28 it serves only questions of pattern β; for unseen β, removing it keeps compatibility with D, so by minimality $C_\beta \notin \Phi$. If for some j no clause of Φ with a literal on x_j is falsified by β, then Φ fails on $(j, \beta \oplus \mathbf{e}_j)$, of pattern β. Otherwise take a clause $C \in \Phi$ falsified by β; $C \neq C_\beta$, so $|C| < m$; take $i \notin \mathrm{supp}(C)$. The question $(i, \beta \oplus \mathbf{e}_i)$ has pattern β, and its positive point $\beta \oplus \mathbf{e}_i$ agrees with β on $\mathrm{supp}(C)$, so it falsifies C, and Φ fails on it (Lemma A.28). Distinct patterns give distinct questions. (c) Each pattern carries m of the $m2^{m-1}$ questions; the rest is the coupon collector's problem, with $\sum_{i \leq N} 1/i = \ln N + 0.5772\ldots + \frac{1}{2N} + O(N^{-2})$. (d) For a seen pattern β, every $\beta \oplus \mathbf{e}_i$ is a seen positive point, so every clause of a compatible Φ is satisfied by all of them (Lemma A.28). A clause falsified by β of width less than m is falsified by $\beta \oplus \mathbf{e}_i$ for i outside its support; so the only admissible clause falsified by β is C_β, and Lemma A.28 requires one. With $R := \{C_\beta : \beta \in \mathrm{pat}(D)\}$ and $\sigma_D := |\mathrm{pat}(D)|$, summing the listing weights of all theories over the base variables that contain R (here $m_{\mathrm{aux}} = 0$) gives, as in the proof of Lemma A.1, at most $\sum_s 2^{-\ell_{\mathbb{N}}(s)} 2^{-\sigma_D \mathrm{len}_m} \sum_{|\Phi'|=s-\sigma_D} \prod_{C \in \Phi'} 2^{-\ell(C)} \leq 2^{-\sigma_D \mathrm{len}_m} \sum_s 2^{-\ell_{\mathbb{N}}(s)}/(s-\sigma_D)! \leq 2^{-\sigma_D \mathrm{len}_m}$. The cumulative loss in the semimeasure convention is $-\log W_{\mathrm{lst}}(D) + \log W_{\mathrm{lst}}(\emptyset)$, and $W_{\mathrm{lst}}(\emptyset) \geq 2^{-c}$. (e) The single equation (respectively the chain encoding) is compatible with every $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, so $W_{\mathrm{lst}} \geq 2^{-\ell_{\mathrm{lst}}}$ of it; the listing lengths are direct from (1). (f) Theorem 1.3(b), (c): the unique usable theory over the base variables has 2^{m-1} clauses of width m, and the chain encoding and the equation are printed from m in polynomial time. □

A simulation (code in the accompanying repository) with uniformly random question sequences for $m = 4, 5, 6$, which computes an exactly ℓ_{lst}-shortest compatible theory at checkpoints, found in each of 330 checkpoints at least as many failures as unseen patterns, as (b) requires, and typically about m times as many; the exact expected times until all patterns are seen are 21.74, 54.09 and 129.87.

Question A.30. Is there a learning theorem of this kind for systems of laws, and for a fixed number of rounds with intermediate concepts? Is there a Bayes lower bound like (d) for arbitrary question sequences that are not block-complete?

A.8 Depth against size for the implication chain

Let $\mathrm{Ch}_m = \{x : x_1 \le \cdots \le x_m\}$, $m \ge 2$. A *k-hop transitive-closure spanner* of the directed path on $[m]$ is a set E of pairs (i, j), $i < j$, such that for all $i < j$ there is a path from i to j in E with at most k edges; $\mathrm{sp}_k(m)$ is the least size of one.

Theorem A.31 (Depth against size: the reduction to spanners)**.** *(i) The prime implicates of Ch_m are exactly the clauses $\neg x_i \vee x_j$, $i < j$. (ii) A 2-CNF of prime implicates, with edge set E, is UP_k-usable iff E is a k-hop transitive-closure spanner of the path. (iii) If an explicit theory Φ over the base variables, with clauses of any width, is UP_k-usable, then for every choice of a prime implicate contained in each clause, the chosen edges form a k-hop transitive-closure spanner of the path. (iv) Hence the least number of clauses of a UP_k-usable theory over the base variables is exactly $\mathrm{sp}_k(m)$, for every k.*

Proof. (i) A partial assignment is consistent with Ch_m iff it sets no $x_i = 1$, $x_j = 0$ with $i < j$; so a clause is an implicate iff it contains some $\neg x_i \vee x_j$ with $i < j$, and these 2-clauses are the prime implicates. (ii) For an edge set E, UP_k from a consistent α derives x_j iff some x_i true in α has a path to j of at most k edges, and $\neg x_i$ iff some x_j false in α is reached from i within k edges (the same clauses read contrapositively); and an entailed literal is entailed by a single literal of α. So usability is the spanner property (take $\alpha = \{x_i\}$ for necessity). (iii) Lemma A.9 holds for every law: the replaced theory derives, round for round, at least what Φ derives, so it is UP_k-usable, and it is a 2-CNF of prime implicates, so (ii) applies (two clauses may give the same edge). (iv) By (iii) a usable Φ has $|\Phi| \ge |E| \ge \mathrm{sp}_k(m)$, and a minimum spanner written as a 2-CNF is usable by (ii). □

The converse of (iii) fails for wide clauses. For $m = 3$, $k = 1$, the theory $\{\neg x_1 \vee x_2,\ \neg x_2 \vee x_3,\ \neg x_1 \vee \neg x_2 \vee x_3\}$ consists of implicates, and choosing $\neg x_1 \vee x_3$ inside the wide clause gives the 1-hop spanner $\{12, 23, 13\}$, but from $x_1 = 1$ one round does not derive x_3. The failure persists when the choice is forced: in $\{\neg x_1 \vee x_2,\ \neg x_1 \vee x_3,\ \neg x_2 \vee x_1 \vee x_3\}$ the wide clause contains only $\neg x_2 \vee x_3$, the edges $\{12, 13, 23\}$ form a 1-hop spanner, but from $x_3 = 0$ one round does not derive $x_2 = 0$. A wide clause propagates in fewer situations than the edge it contains.

The minimum size of a k-hop transitive-closure spanner of the directed path is $\Theta(m\,\mathrm{ia}_k(m))$, with ia_k the k-th row of the inverse Ackermann function: $\mathrm{ia}_2(m) = \log m$, $\mathrm{ia}_3(m) = \log\log m$, $\mathrm{ia}_4(m) = \log^* m$. This is the bound of Alon and Schieber (1987) for answering product queries in at most k steps, in the language of transitive-closure spanners (Bhattacharyya et al., 2012; Berman et al., 2014); Yao (1982) gave the first inverse-Ackermann trade-off for range queries. A direct computation (integer programming, confirmed by exhaustive search for $m \le 7$; code in the accompanying repository) gives the exact values:

m	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\mathrm{sp}_1(m) = \binom{m}{2}$	1	3	6	10	15	21	28	36	45	55	66	78	91	105	120	136	153
$\mathrm{sp}_2(m)$	1	2	4	6	8	10	13	16	19	22	25	28	31	34	38	42	46
$\mathrm{sp}_3(m)$	1	2	3	5	6	8	10	12	14	16	18	20					
$\mathrm{sp}_4(m)$	1	2	3	4	6	7	9	10	12	14	16						

At these sizes $\mathrm{sp}_3(m) = 2m - 6$ for $6 \le m \le 13$; the $\log\log$ growth is not yet visible.

Proposition A.32 (The midpoint construction)**.** *The midpoint recursion, $S(1) = 0$ and $S(n) = n - 1 + S(\lceil \frac{n-1}{2} \rceil) + S(\lfloor \frac{n-1}{2} \rfloor)$, gives 2-hop spanners of the path, so $\mathrm{sp}_2(m) \le S(m) = \sum_{i=1}^{m} \lfloor \log i \rfloor = (m+1)L - 2^{L+1} + 2$, where $L := \lfloor \log m \rfloor$.*

Proof. Join a hub vertex h to all other vertices ($i \to h$ for $i < h$, $h \to j$ for $j > h$), with $\lceil \frac{n-1}{2} \rceil$ vertices on one side and $\lfloor \frac{n-1}{2} \rfloor$ on the other, and recurse on both sides: pairs on opposite sides of h, or containing h, are joined within 2 hops, and pairs on one side by the recursion. $\Sigma(m) := \sum_{i \le m} \lfloor \log i \rfloor$ satisfies the same recursion: splitting $i = 2i'$ and $i = 2i' + 1$ ($i \ge 2$), $\lfloor \log i \rfloor = 1 + \lfloor \log i' \rfloor$, so $\Sigma(m) = (m-1) + \Sigma(\lfloor m/2 \rfloor) + \Sigma(\lfloor (m-1)/2 \rfloor)$, and $\lfloor m/2 \rfloor = \lceil (m-1)/2 \rceil$. The closed form is standard. □

The construction attains $\mathrm{sp}_2(m)$ for every $m \le 18$ in the table above.

Conjecture A.33. $\mathrm{sp}_2(m) = \sum_{i=1}^{m} \lfloor \log i \rfloor$ *for every* $m \ge 2$.

We have neither a proof nor a counterexample; a computation confirms it for $m \le 18$. The best lower bound we prove is that of Theorem A.34, about $\frac{1}{2} m \log m$, which holds even with intermediate concepts.

Theorem A.34 (Intermediate concepts and wide clauses gain at most a factor about 2 at depth 2)**.** *Let* $m \ge 4$. *Every* UP_2*-usable explicit theory* Φ *for* Ch_m*, with any auxiliary variables and clauses of any width, has*

$$|\Phi| \;\ge\; \tfrac{1}{2}\Big((m-2)\log \tfrac{m-2}{2} - m\Big),$$

while $\mathrm{sp}_2(m) \le m \log m$. *So at depth* 2 *the least usable theory has* $\Theta(m \log m)$ *clauses with or without intermediate concepts, and they save at most a factor of about* 2.

Proof. Upper bound. In the midpoint recursion $S(n) = n - 1 + S(a) + S(b)$ with $a + b = n - 1$ and $a, b \le n/2$, so by induction $S(n) \le n - 1 + (n-1)\log(n/2) \le n \log n$. *Simplification.* Let U_0 be the set of literals derived by unit propagation from Φ alone; they are auxiliary, as Ch_m fixes no base variable. Simplifying Φ by U_0 (deleting the clauses satisfied by U_0 and the literals falsified by it) gives Φ' with $|\Phi'| \le |\Phi|$ and the same projection, which does not mention the variables of U_0 and has no unit clauses, and in which every UP_2 derivation of Φ outside U_0 is made no later (induction on rounds: a clause of Φ that fires in a run from a base antecedent contains no literal of U_0, since such a literal would be falsified, that is, its negation derived, against soundness; so its simplification is in Φ' and fires). So we may assume that Φ has no unit clauses. *Single sources.* For a base index v let $\mathrm{Nb}(v) := \{x_v\} \cup \{L : (\neg x_v \vee L) \in \Phi\}$, the literals known after one round from $\alpha = \{x_v\}$. For a clause C containing x_j let $P(C, j) := \{\neg L' : L' \in C \setminus \{x_j\}\}$, its premises for deriving x_j. From $\{x_u\}$, UP_2 derives x_j iff some $C \ni x_j$ has $P(C, j) \subseteq \mathrm{Nb}(u)$ (round 1 is the case $C = \neg x_u \vee x_j$); say that C *serves* u for j. By soundness, if C serves u for j then $u \le j$. A clause serving any source for j contains no other positive base literal $x_{w'}$, since its premise $\neg x_{w'}$ would follow from a single x_u, which is unsound. Let $\mathrm{cnt}(j)$ be the number of clauses containing x_j that serve some $u < j$; then $\sum_j \mathrm{cnt}(j) \le |\Phi|$, and $\sum_v |\mathrm{Nb}(v)| \le m + |\Phi|$, since a 2-clause $\neg x_v \vee L$ is counted only for v (a 2-clause with two negative base literals is not an implicate). *Random experiment.* Give every literal an independent fair coin, *left* or *right*. Call v *source-alive* if all of $\mathrm{Nb}(v)$ is right, and j *target-alive* if every clause containing x_j that serves some $u < j$ has a left premise. No $u < j$ has u source-alive and j target-alive: by usability some C serves u for j, and all its premises lie in $\mathrm{Nb}(u)$ and are right. *Pairing.* For $a = 1, \dots, M' := \lfloor m/2 \rfloor$ let $\mathcal{B}_a$ be the event that $2a$ is source-alive and $2a - 1$ target-alive. For $a < a'$ we have $2a < 2a' - 1$, so at most one $\mathcal{B}_a$ occurs and $\sum_a \Pr[\mathcal{B}_a] \le 1$. Given that $\mathrm{Nb}(2a)$ is all right, a clause serving some $u < 2a - 1$ for $2a - 1$ cannot have all its premises in $\mathrm{Nb}(2a)$ (it would serve $2a$ for $2a - 1$, which is unsound), so it is blocked by a left coin outside $\mathrm{Nb}(2a)$ with probability at least $\frac{1}{2}$, independently of the coins of $\mathrm{Nb}(2a)$; these blocking events are increasing in the set of left coins, so by Harris's inequality

they occur together with probability at least $2^{-\mathrm{cnt}(2a-1)}$. Hence $\Pr[\mathcal{B}_a] \geq 2^{-|\mathrm{Nb}(2a)|-\mathrm{cnt}(2a-1)}$, and by the convexity of $x \mapsto 2^{-x}$, $\sum_a(|\mathrm{Nb}(2a)| + \mathrm{cnt}(2a-1)) \geq M' \log M'$. The shifted pairing ($2a+1$ source-alive and $2a$ target-alive, for $a \leq M'' := \lfloor (m-1)/2 \rfloor$; $a < a'$ gives $2a+1 < 2a'$) gives $\sum_a(|\mathrm{Nb}(2a+1)| + \mathrm{cnt}(2a)) \geq M'' \log M''$. Adding, and using $M', M'' \geq (m-2)/2 \geq 1$, $m + 2|\Phi| \geq \sum_v |\mathrm{Nb}(v)| + \sum_j \mathrm{cnt}(j) \geq (m-2) \log \frac{m-2}{2}$. □

The bound is asymptotic: it is positive only from $m = 8$ on, and gives 11.7 at $m = 16$ against $\mathrm{sp}_2(16) = 38$. The argument adapts Hansel's lemma, that covering the complete graph on n vertices by bicliques needs total size at least $n \log n$ (Hansel, 1964; Katona and Szemerédi, 1967), to ordered pairs and to conjunctive premises: the pairing keeps the diagonal out, and Harris's inequality handles clauses with several premises. For 2-CNFs, after the simplification step the implication graph of a UP_2-usable theory (edges $\neg a \to b$ and $\neg b \to a$ for each clause $a \vee b$) is a Steiner 2-hop transitive-closure spanner of the path, with the positive base literals as its vertices and at most $2|\Phi|$ edges, so lower bounds for Steiner 2-transitive-closure spanners of directed hypergrids (Berman et al., 2014), of which the path is the one-dimensional case, bear on that case; what Theorem A.34 adds is the case of wide clauses and an explicit constant.

Proposition A.35 (Every fixed depth: superlinear with intermediate concepts)**.** *Suppose that for every fixed depth d, every monotone circuit of depth d with unbounded fan-in AND and OR gates that computes all the prefix ORs x_1, $x_1 \vee x_2$, …, $x_1 \vee \cdots \vee x_m$ has $m \cdot \omega(1)$ wires. Then for every fixed k and every bound w on clause width, every UP_k-usable theory Φ for Ch_m with clauses of width at most w, with any auxiliary variables, has $\|\Phi\| \geq m \cdot \omega(1)$.*

Lower bounds of this form for constant-depth prefix circuits are due to Chandra, Fortune and Lipton, in a companion paper to Chandra et al. (1985), which gives the matching constant-depth constructions; we state the property as a hypothesis. Size must be counted in wires: counting gates, prefix OR has depth-1 circuits with m gates.

Proof. On the antecedents $\alpha_{A'} = \{x_i : i \in A'\}$, $A' \subseteq [m]$, all consistent, soundness and usability give: UP_k derives x_j iff $A' \cap [1, j] \neq \emptyset$. So the circuit of Lemma A.18(b), with inputs the literals x_i, is a monotone circuit of depth $2k$ with at most $k(w\|\Phi\| + 2\bar{m})$ wires that computes all prefix ORs, and $\bar{m} \leq m + \|\Phi\|$. Hence $(w+2)k\|\Phi\| + 2km \geq m \cdot \omega(1)$. □

Question A.36. (a) What is the least UP-usable theory over the base variables when $d_* = 2$, beyond equivalence laws? (b) What is the constant at depth 2 with intermediate concepts, between $\frac{1}{2} m \log m$ (Theorem A.34) and $\sum_{i \leq m} \lfloor \log i \rfloor \approx m \log m$ (Conjecture A.33)? What is the effect of intermediate concepts at depth $k \geq 3$ beyond Proposition A.35? (c) How many rounds does one parity law need for usable theories of polynomial size: between $\Omega(\log m / \log \log m)$ and $2\lceil \log m \rceil - 1$ (Proposition 1.8(d))?

A.9 Lemmas in first-order logic

In first-order logic the cost of a lemma-free theory for an analytic reasoner can be non-elementary: there are sentences with proofs of size linear in k in Gentzen's sequent calculus LK with cuts whose cut-free proofs have size not bounded by any Kalmár-elementary function of k (Orevkov, 1982); see also the non-elementary lower bounds of Statman (1979). Cut formulas are lemmas, and cut-free proofs are lemma-free reasoning.

Remark A.37 (Valid lemma sentences restore short cut-free proofs). Let $\mathcal{D}$ be an LK proof of $T_0 \vdash \varphi$, with $T_0 \subseteq T$ finite and cuts numbered $1, \ldots, n_{\mathrm{cut}}$, the i-th with left premise $\Gamma_i \vdash \Delta_i, A_i$.

Let $L_i := \forall \vec{v}\,(\bigwedge \Gamma_i \to \bigvee \Delta_i \vee A_i)$, the universal closure over the free variables. Then (a) each L_i is valid, since the sequent $\Gamma_i \vdash \Delta_i, A_i$ is derivable and LK is sound, so $T \cup \{L_1, \ldots, L_{n_{\mathrm{cut}}}\}$ is deductively equivalent to T; and (b) $L_1, \ldots, L_{n_{\mathrm{cut}}}, T_0 \vdash \varphi$ has a cut-free proof of size polynomial in the size of $\mathcal{D}$.

Proof sketch. We do not write out the bookkeeping of contexts in the chosen sequent calculus. Replace the i-th cut, with premises $\Gamma \vdash \Delta, A$ and $A, \Gamma' \vdash \Delta'$, as follows. From the right premise and identity sequents $\delta \vdash \delta$ ($\delta \in \Delta$), $\vee$-left gives $\bigvee \Delta \vee A, \Gamma' \vdash \Delta, \Delta'$; from identity sequents $\gamma \vdash \gamma$, $\wedge$-right gives $\Gamma \vdash \bigwedge \Gamma$; $\to$-left gives $\bigwedge \Gamma \to \bigvee \Delta \vee A, \Gamma, \Gamma' \vdash \Delta, \Delta'$; and $\forall$-left steps, instantiating the bound variables by the free variables of the cut's premises, give $L_i, \Gamma, \Gamma' \vdash \Delta, \Delta'$. The subproof of the left premise is dropped. Carry L_i down to the end sequent, by weakening where branches meet and contraction for duplicates; L_i is closed, so no eigenvariable condition below the former cut is affected. Identity sequents for composite formulas have cut-free proofs of size polynomial in the formula. □

The added sentences are valid: the theory's deductive closure, and so everything the idealised inducer sees, is unchanged, and only its written form changes. If every cut's left premise has the form $T_i \vdash A_i$ with $T_i \subseteq T$, the closures $\forall \vec{v}\, A_i$ of the cut formulas are theorems of T and serve directly; in general they are not, since a cut formula can be arbitrary.

Proposition A.38 (When a lemma pays)**.** *Let* $\mathrm{Obj}(T) := \mathrm{lh}(T) + \sum_{\varphi \in D} \rho_T(\varphi)$*, with* lh *additive over axioms and* $\rho_T(\varphi)$ *the cost of the cheapest derivation of the datum* φ *from* T*, non-increasing when axioms are added. Adding a theorem* $L \notin T$ *of* T *as an axiom lowers* Obj *if and only if* $\sum_{\varphi \in D}(\rho_T(\varphi) - \rho_{T \cup \{L\}}(\varphi)) > \mathrm{lh}(L)$*: a lemma enters iff the derivation cost it saves on the data exceeds its own length.*

Proof. By additivity, $\mathrm{Obj}(T \cup \{L\}) - \mathrm{Obj}(T) = \mathrm{lh}(L) - \sum_{\varphi \in D}(\rho_T(\varphi) - \rho_{T \cup \{L\}}(\varphi))$. □

This is the general form of the observation in Remark E.9(iv) of Hänni and Claude (Anthropic) (2026) that a time-bounded likelihood favours adopting theorems with long proofs as axioms.

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